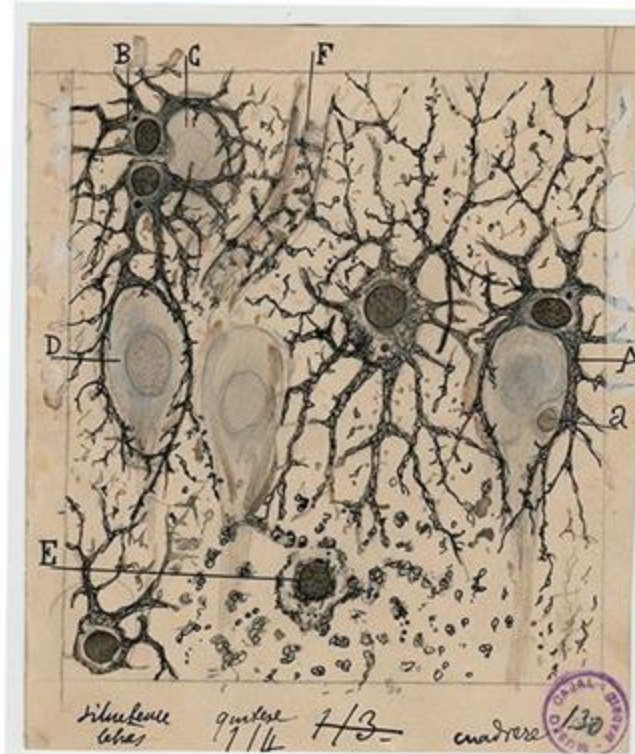


Single cell RNA-seq, part I

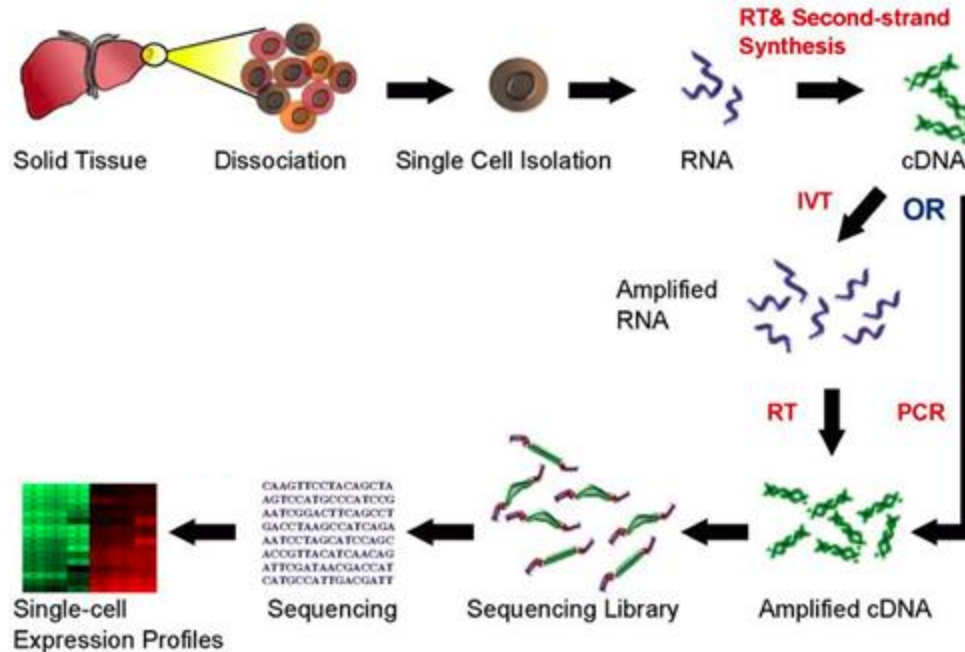
Martin Hemberg
STAT115/215, BIO/BST282
March 23, 2021

Shortcomings of bulk RNA-seq

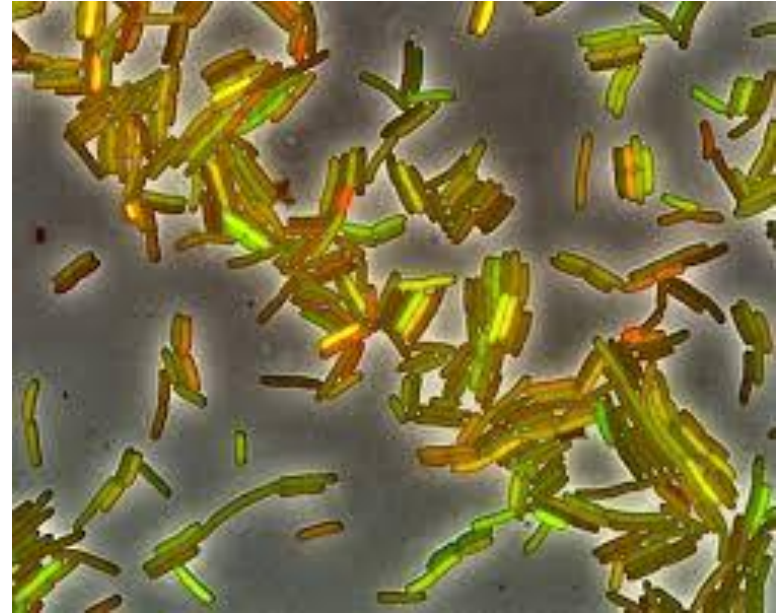
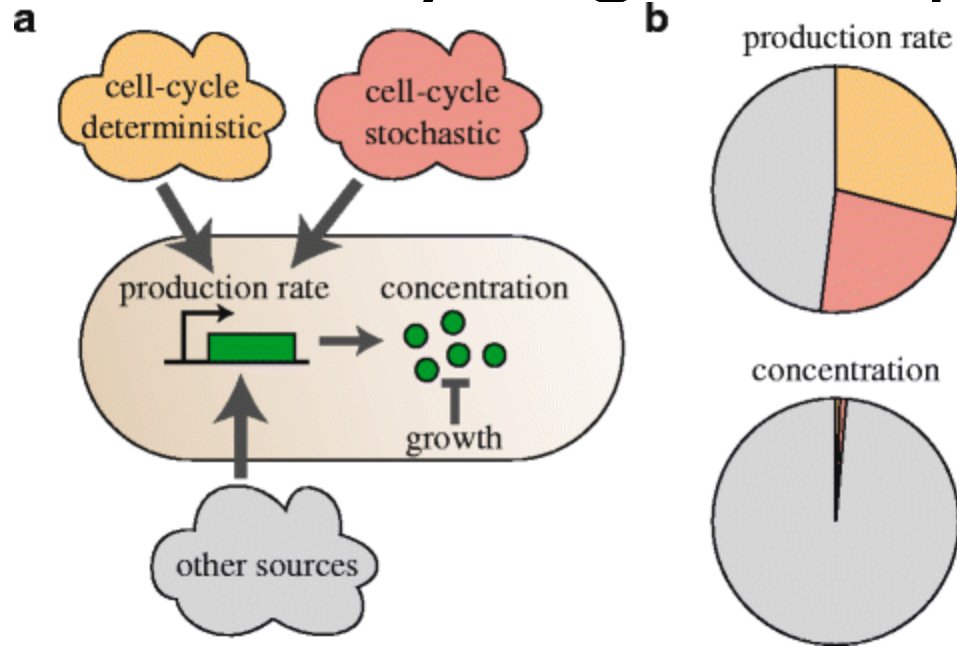


Single-cell workflow

Single Cell RNA Sequencing Workflow



Single-cell RNA-seq captures variability in gene expression



The Human Cell Atlas



HUMAN
**CELL
ATLAS**

Cell atlases for other organisms

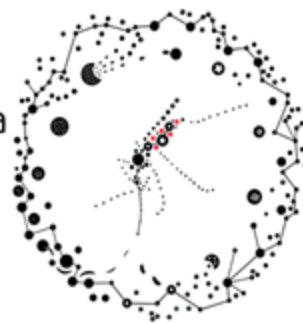


**HUMAN
CELL
ATLAS**

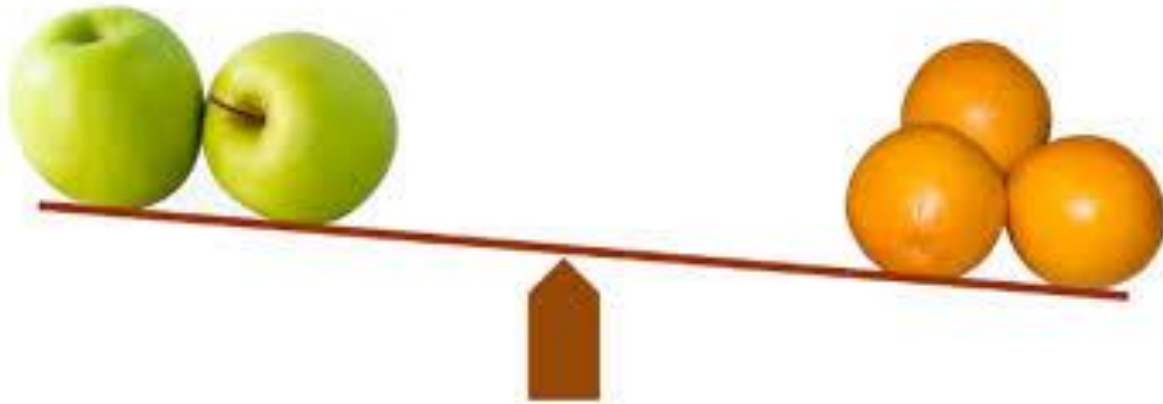
The Mouse Cell Atlas



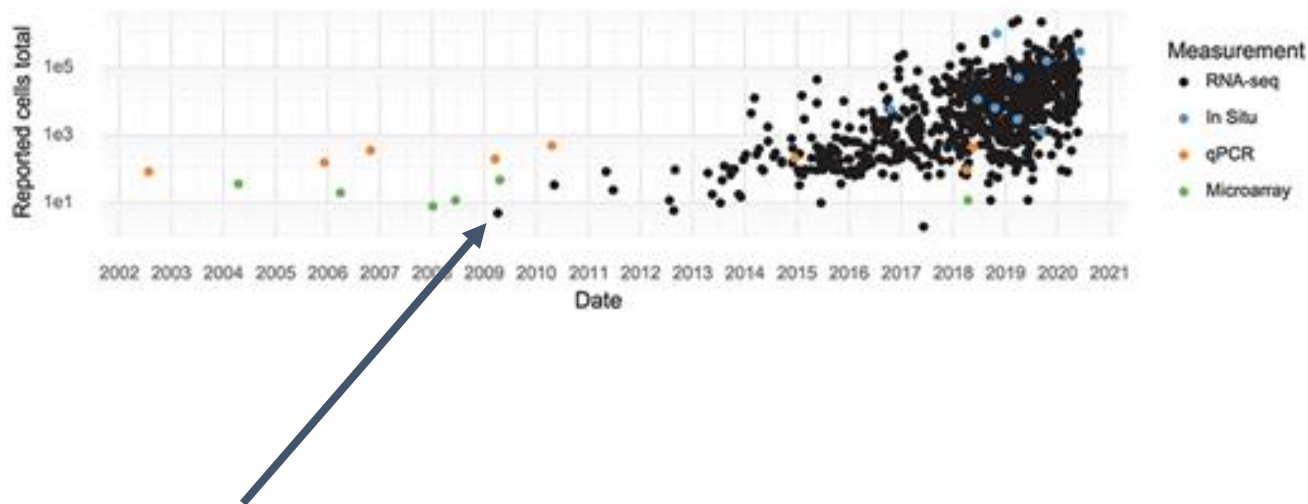
Malaria
Cell
Atlas



When is bulk RNA-seq “better”?

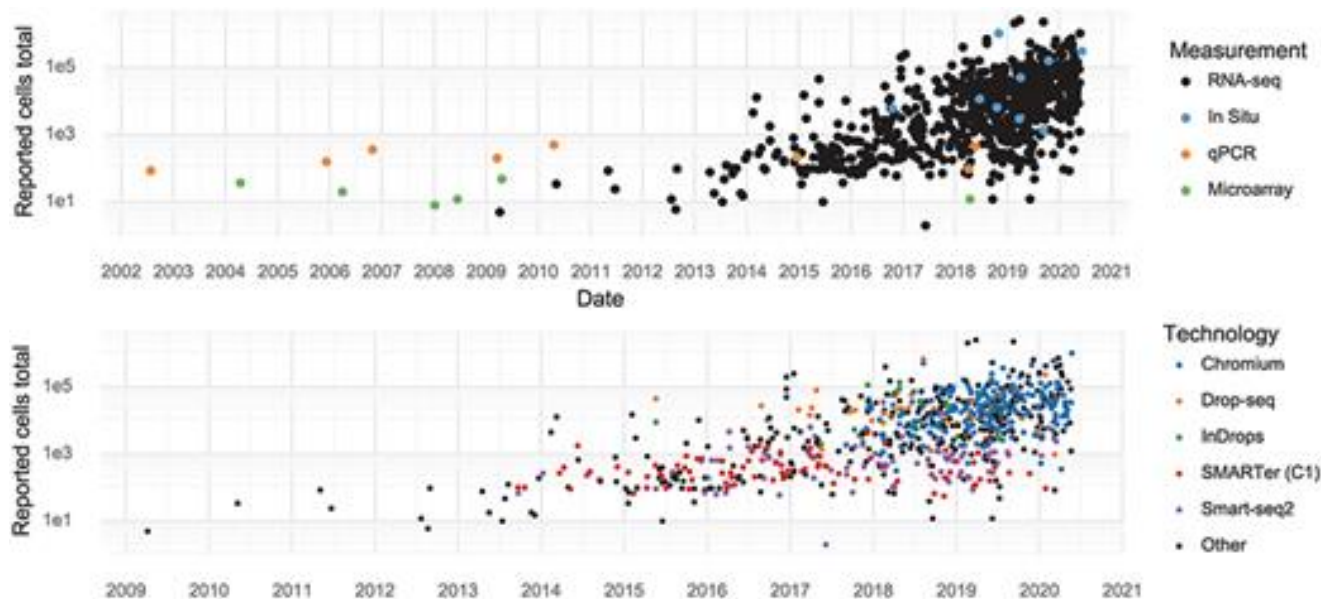


Exponential growth of scRNA-seq



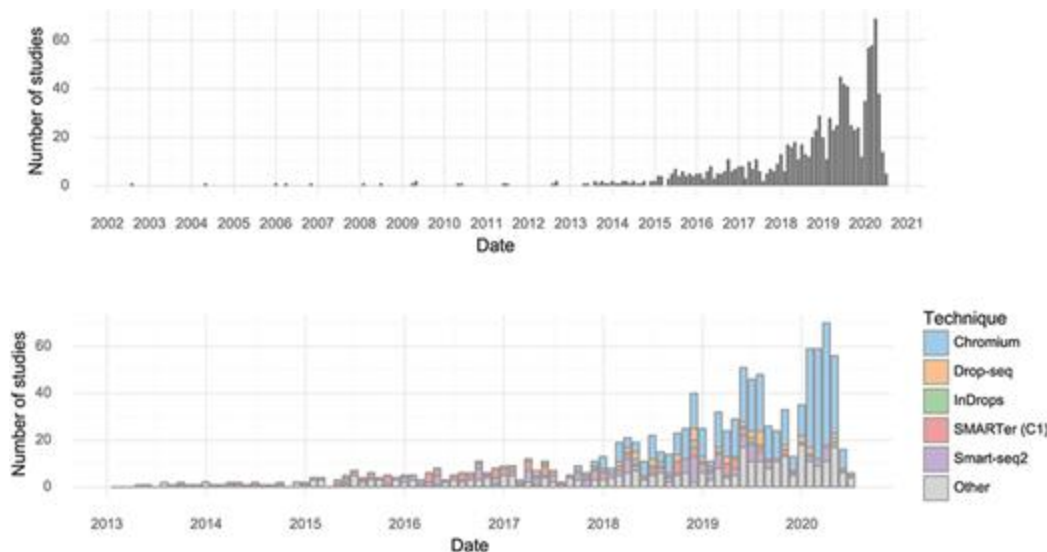
Svensson et al,
2020, Databases

Exponential growth of scRNA-seq



Svensson et al,
2020, Databases

Exponential growth of scRNA-seq



Svensson et al,
2020, Databases

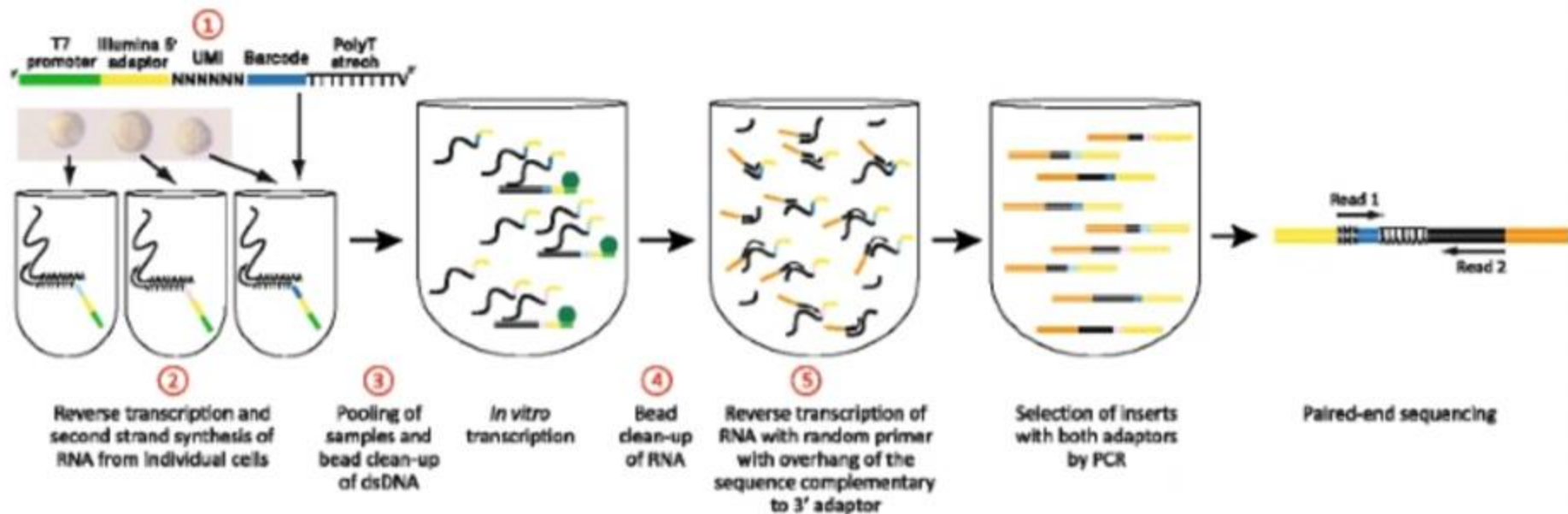
scRNA-seq technologies

Smart-seq2

- SMART - Switching Mechanism At the end of the 5'-end of the RNA Transcript

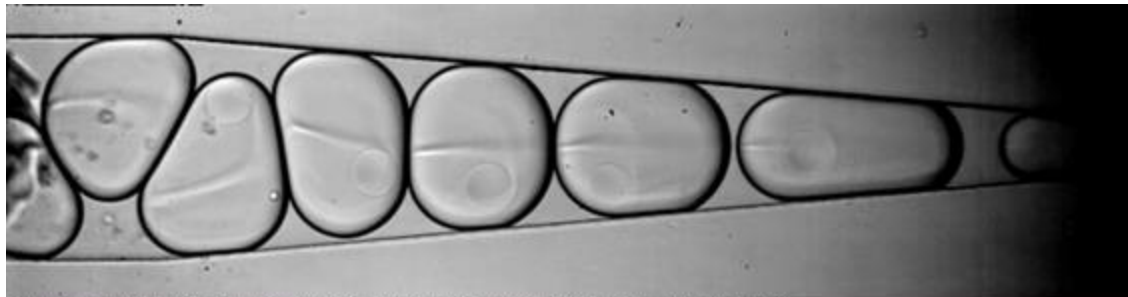


CEL-seq2

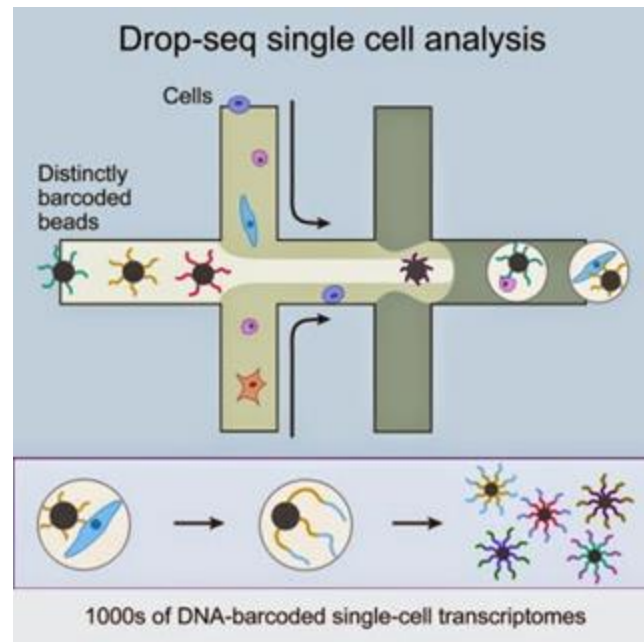


Hashimshony et al,
Genome Biology, 2016

Drop-seq and InDrop

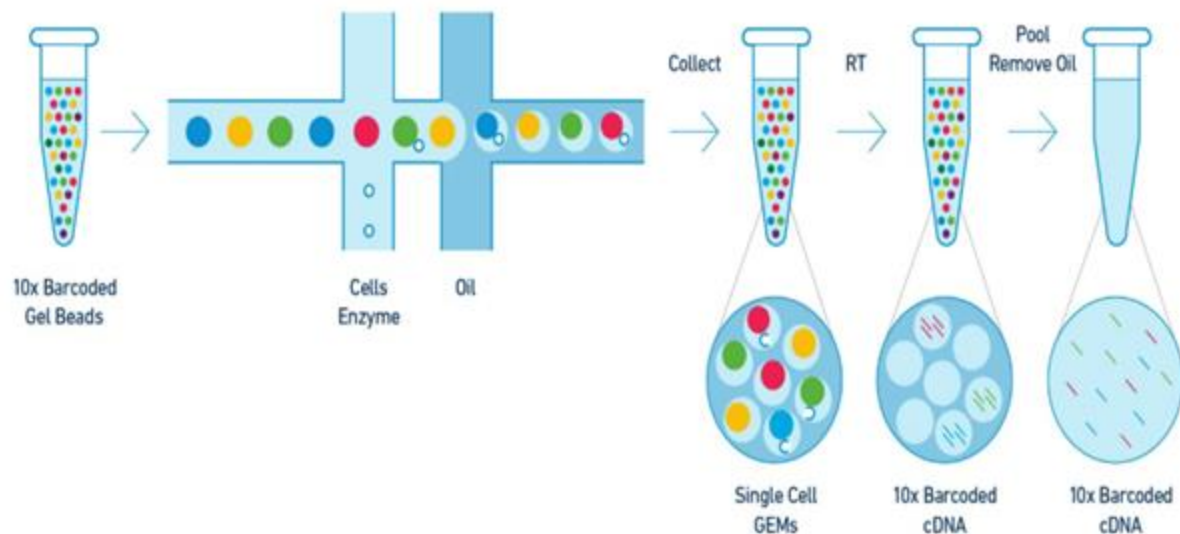


6/3/2014 9:06:59 PM -53518.4[ms] 000003892 HiSpec 1 [00-11-1c-f1-73-f3] Fastec 1280x336(Q) 300fps 100µs V1.4.3 (Build: 2419)



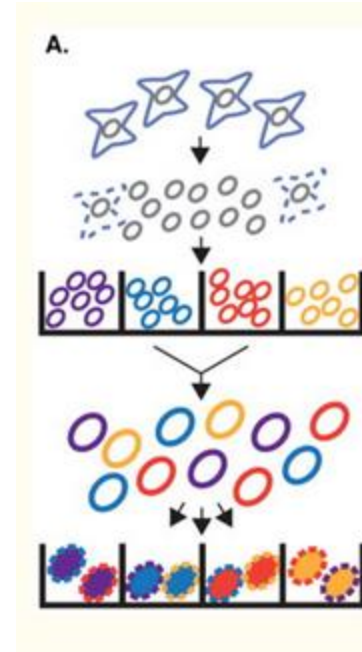
Klein et al,
Macosko et al,
Cell, 2016

10X Genomics (Chromium)



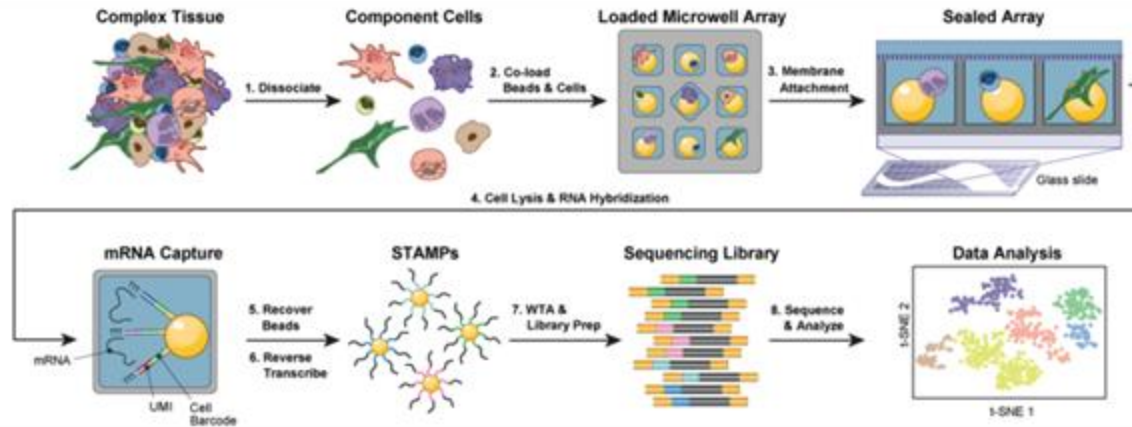
Zhent et al, Nat
Communications, 2017

Single-cell Combinatorial Indexing



Cusanovich et al,
Science, 2015

Seq-well uses microwells

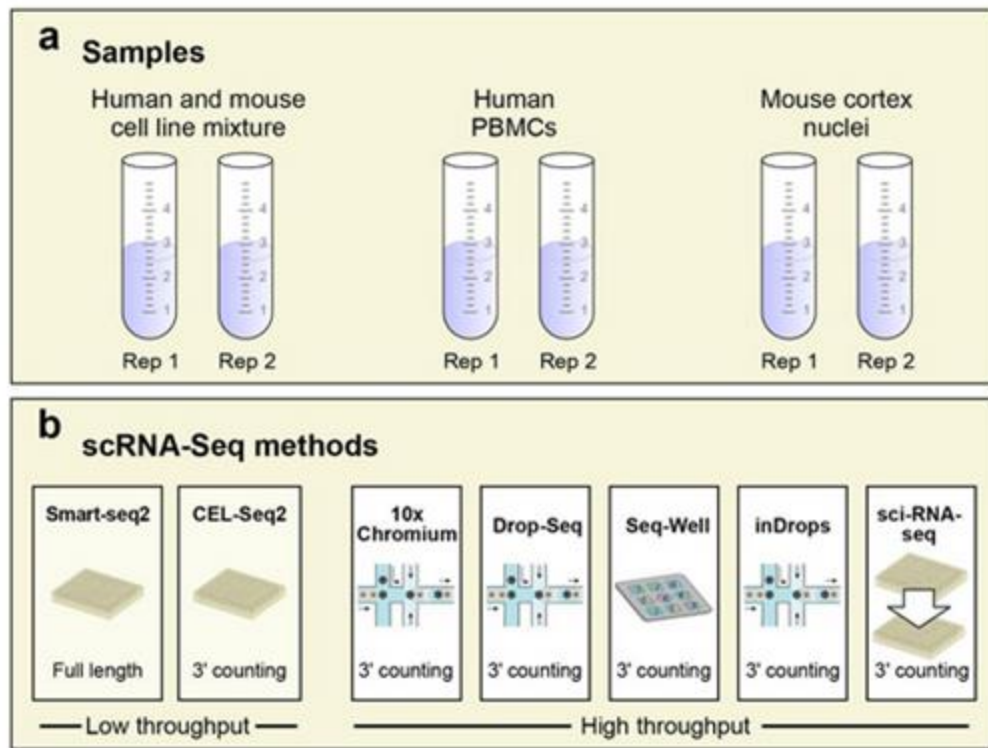


Geirahn et al, Nat
Methods, 2017

Choosing a protocol

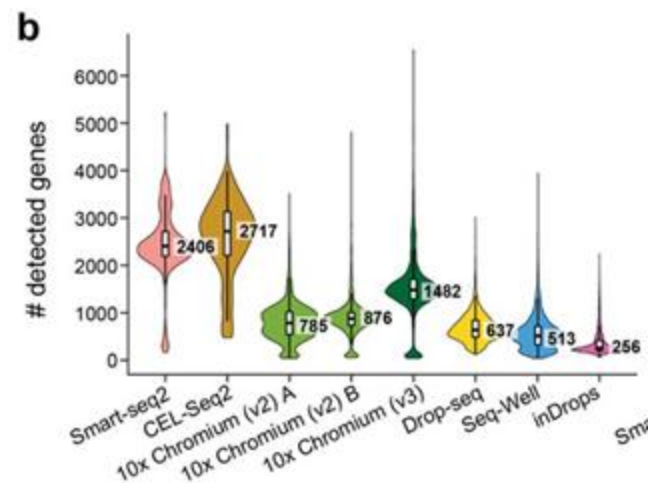
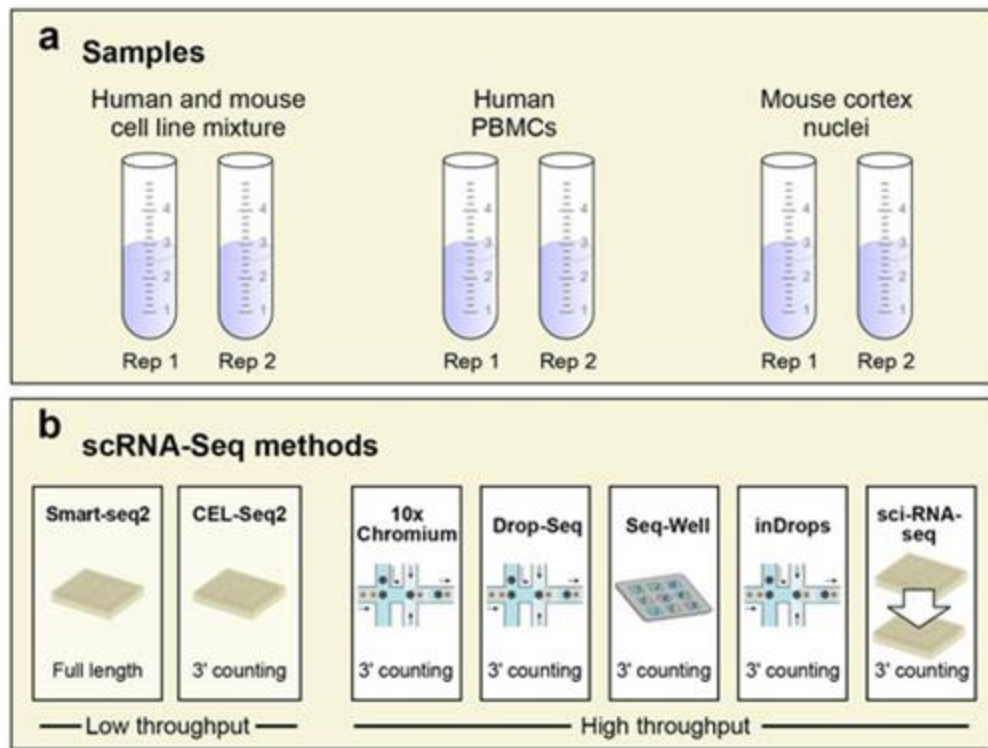
- Droplet based technologies have high throughput
- Droplet based methods are less sensitive and will detect fewer genes
- Initially, droplet based methods would only sequence the 3'-end of mRNAs, but now the standard is to sequence the 5'-end

Formal benchmarks



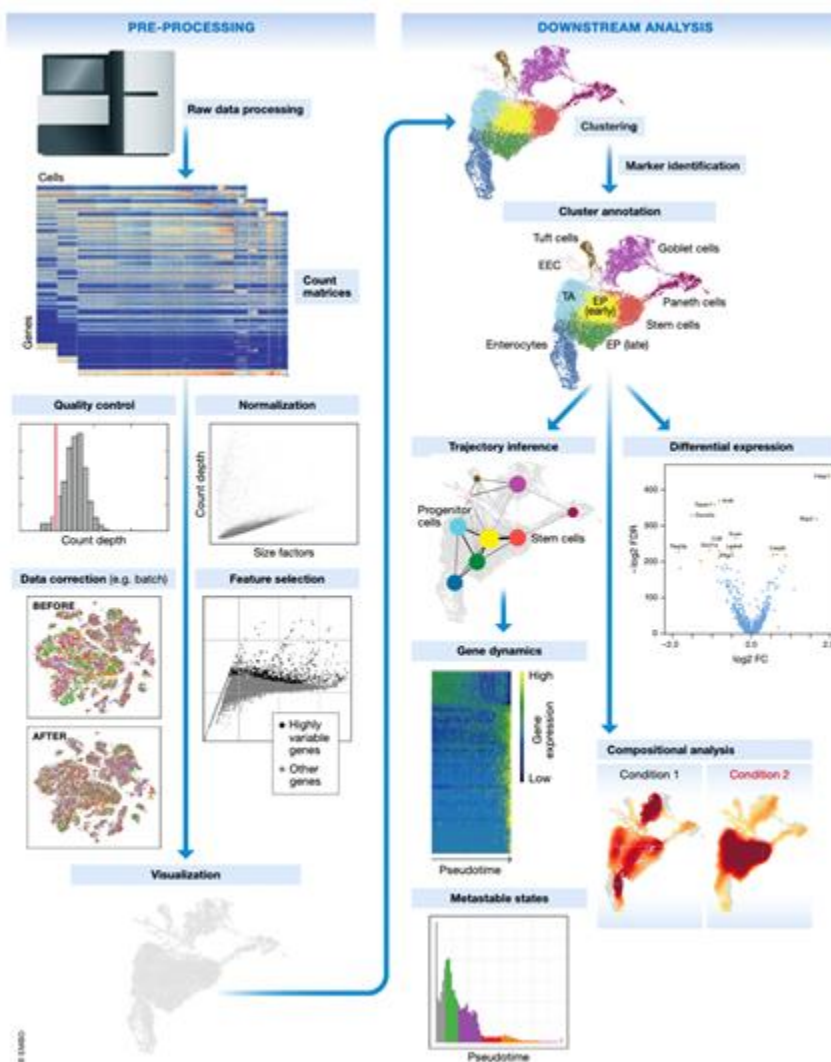
Ding et al, Nat
Biotech, 2020

Formal benchmarks



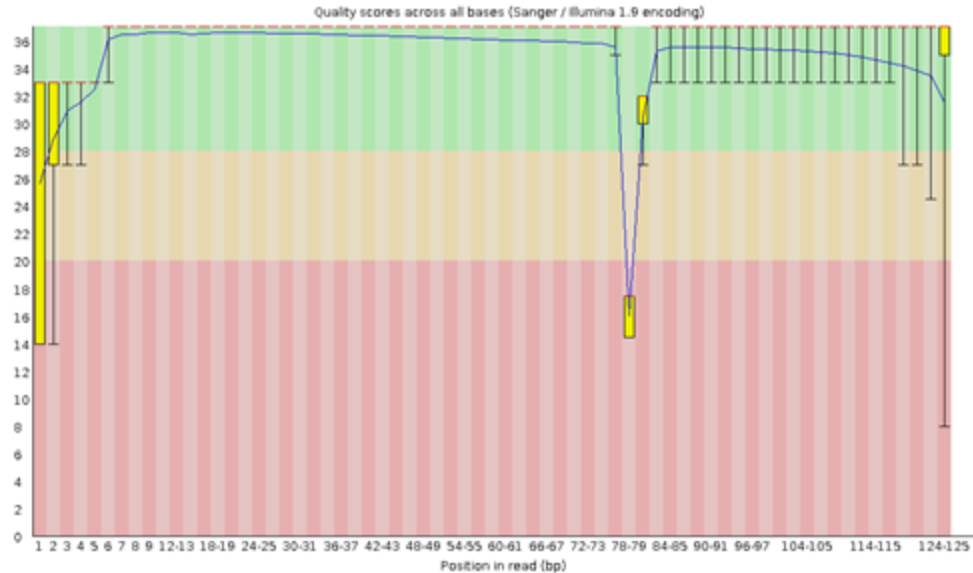
Ding et al, Nat
Biotech, 2020

Preprocessing and quality control

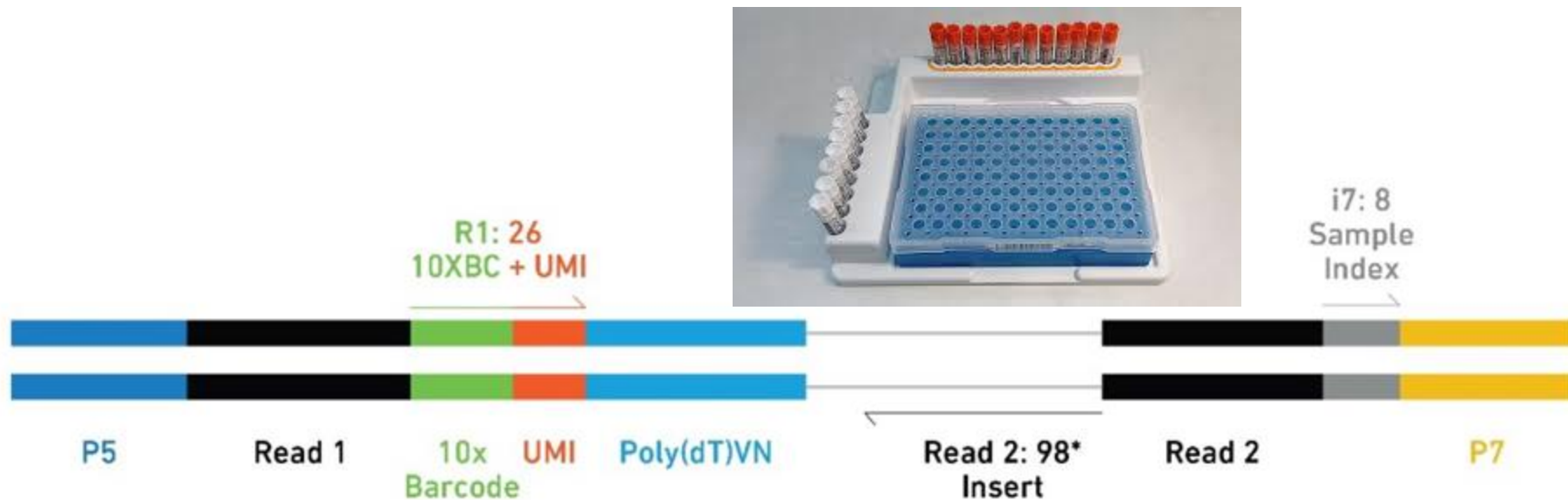


Luecken and Theis,
MSB, 2019

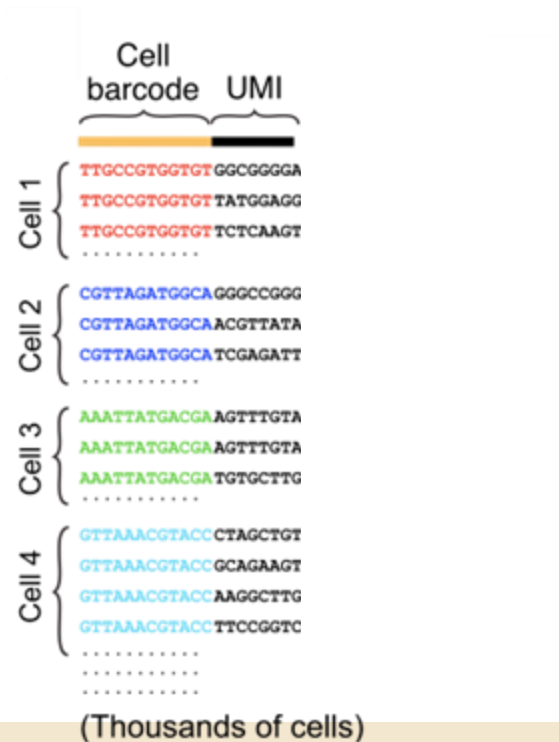
Reads QC and trimming



Demultiplexing identifies cells

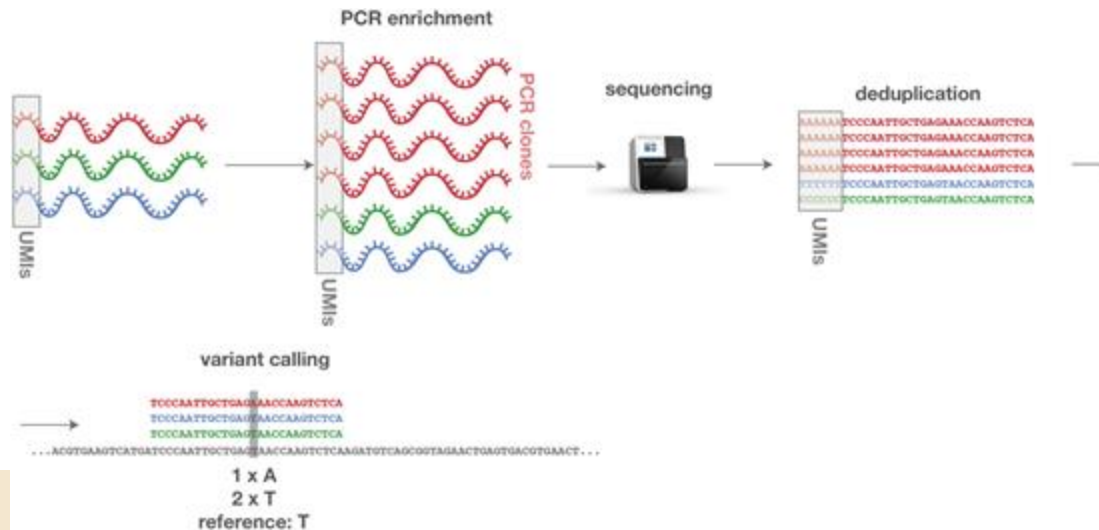


Overview of barcodes

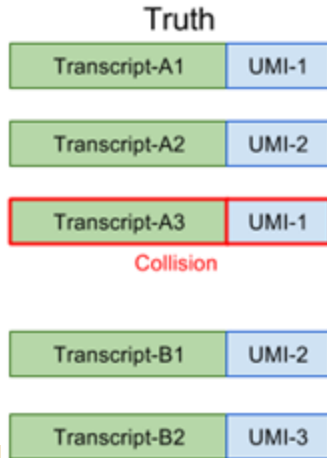


Unique Molecular Identifiers (UMIs)

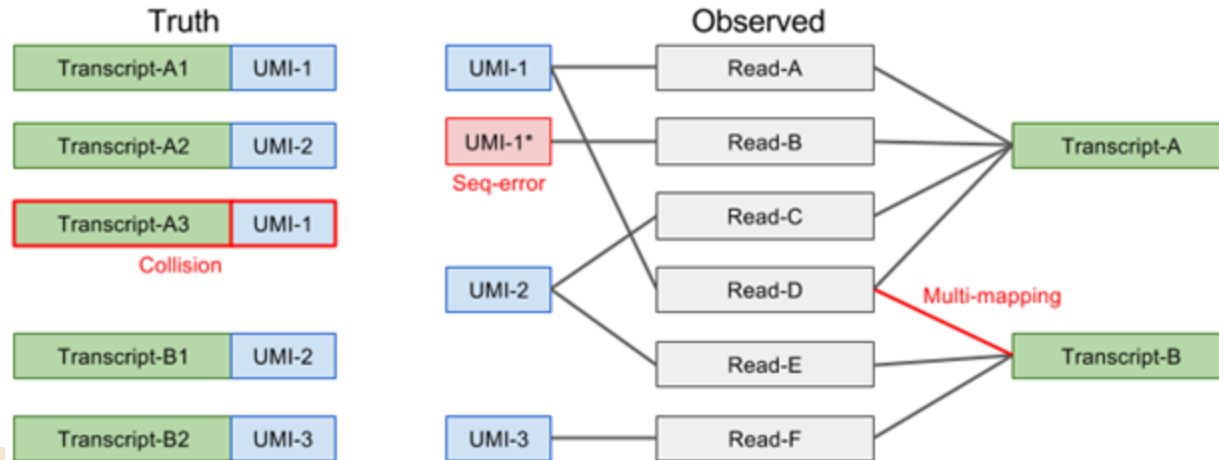
- Tag-based protocols allow for UMIs to improve quantification
 - Remove PCR duplicates



Unique Molecular Identifiers (UMIs)



Unique Molecular Identifiers (UMIs)



UMI-tools corrects using a graph

E

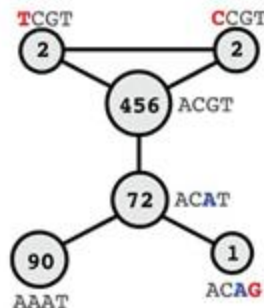
Unique (6)

ACGT TCGT
ACAG CCGT
ACAT AAAT

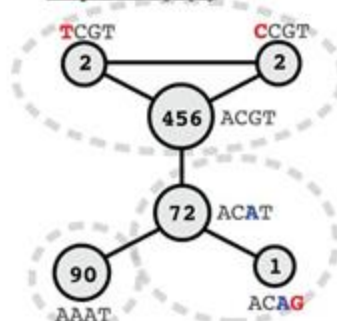
Percentile (5)

ACGT: 456 TCGT: 2 Mean = 104
AAAT: 90 CCGT: 2 Threshold = 1.04
ACAT: 72 ACAG: 1 DROP: ACAG

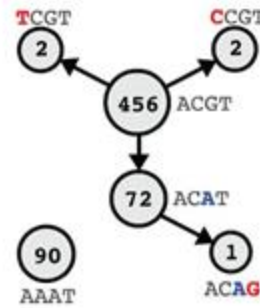
Cluster (1)



Adjacency (3)

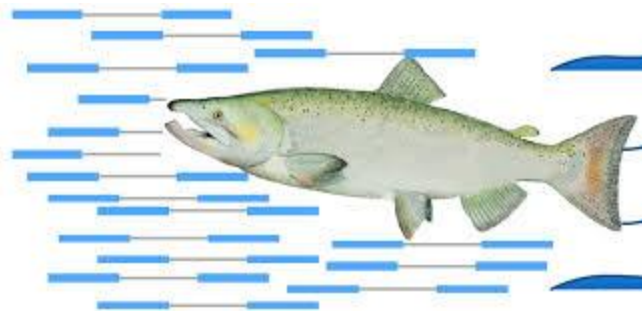


Directional (2)

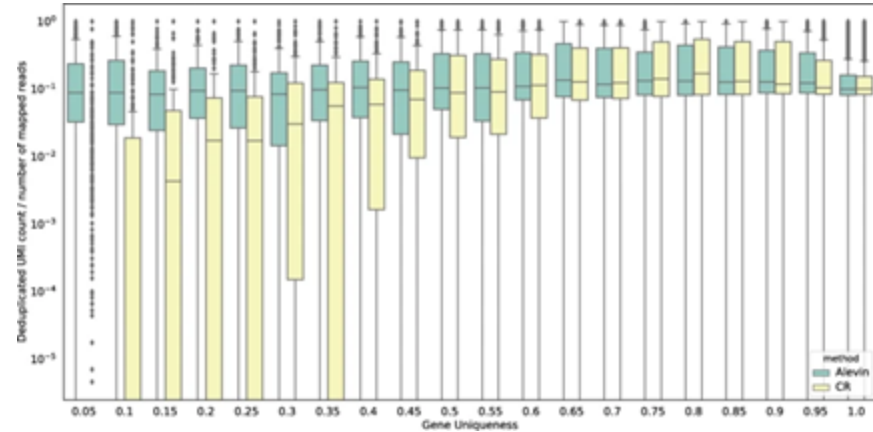
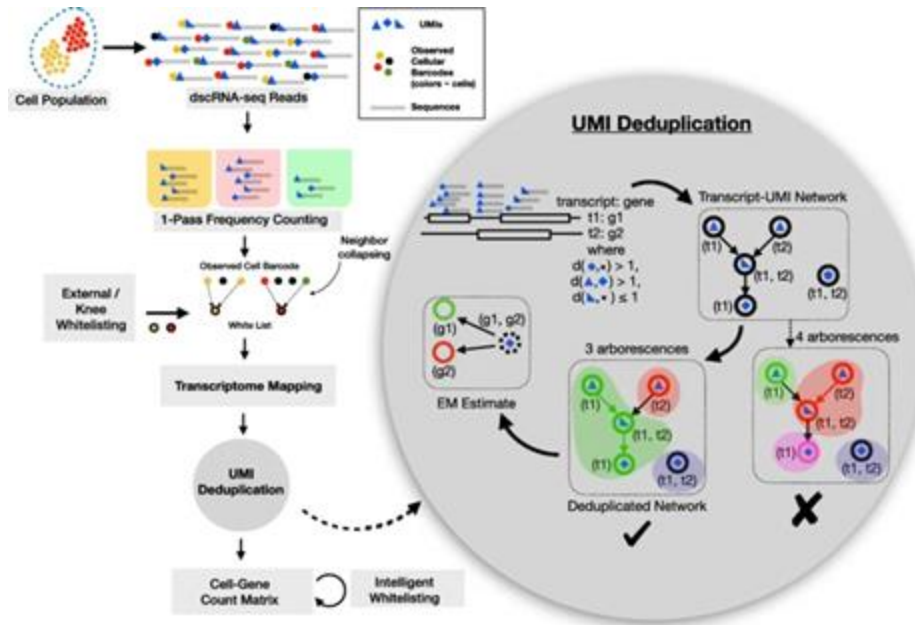


Smith et al, Genome Research, 2017

Aligning reads and quantification



Combined UMI analysis and quantification using Alevin



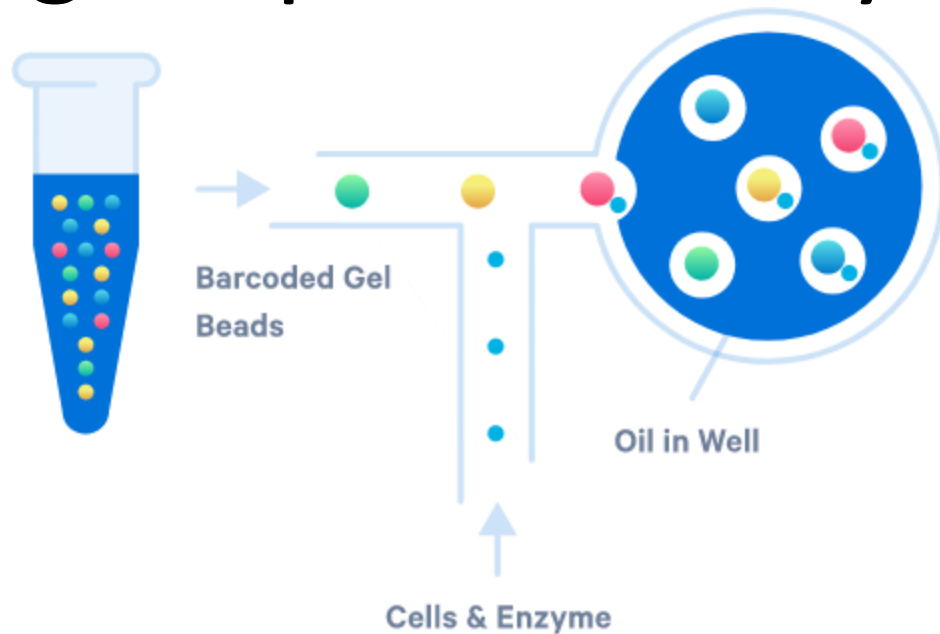
Srivastava et al,
Genome Biology, 2019

Overview of aligned reads

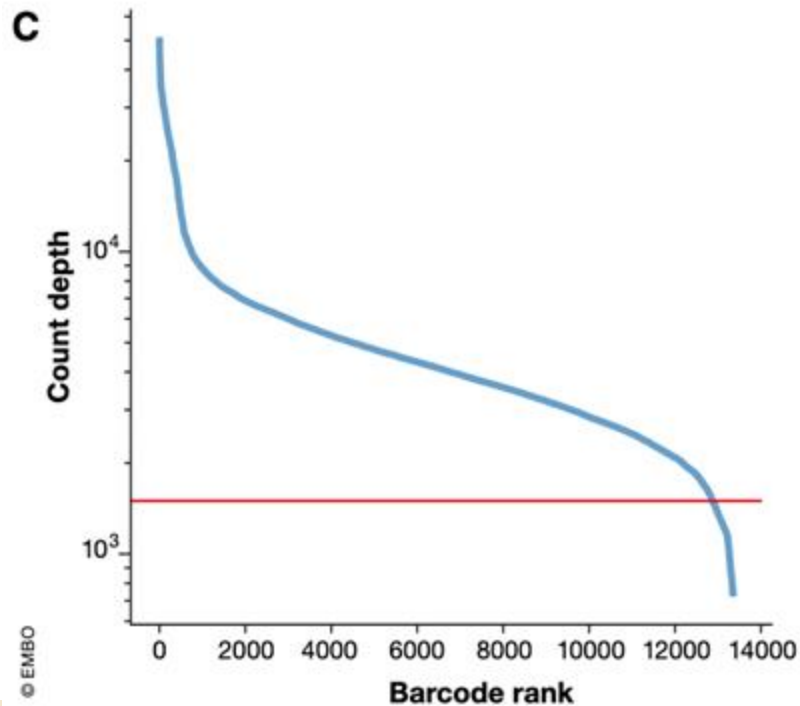
	Cell barcode	UMI	cDNA		
					
Cell 1	TTGCCGTGGTGT	GGCGGGGA	CGGTGTTA	DDX51	1
		TATGGAGG	CCAGCACC	NOP2	1
		TCTCAAAGT	AAAATGGC	ACTB	1
					
Cell 2	CGTTAGATGGCA	GGGCCGGG	CTCATAGT	LBR	1
		ACGTTATA	ACGCGTAC	ODF2	1
		TCGAGATT	AGCCCTTT	HIF1A	1
					
Cell 3	AAATTATGACGA	AGTTTGTA	GGGAATTA	ACTB	2
		AGTTTGTA	AGATGGGG		
		TGTGCTTG	GACTGCAC	RPS15	1
					
Cell 4	GTTAAACGTACC	CTAGCTGT	GATTTTCT	GTPBP4	1
		GCAGAAAGT	GTTGGCGT	GAPDH	1
		AAGGCTTG	CAAAGTTC	ARL1	2
		TTCCGGTC	TCCAGTCG		
					
					
					

(Thousands of cells)

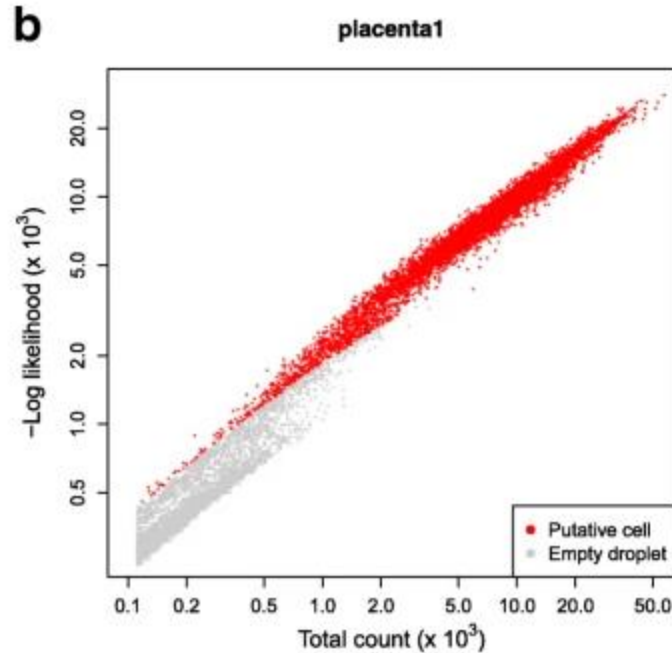
Loading droplet based systems



Empty droplets detection

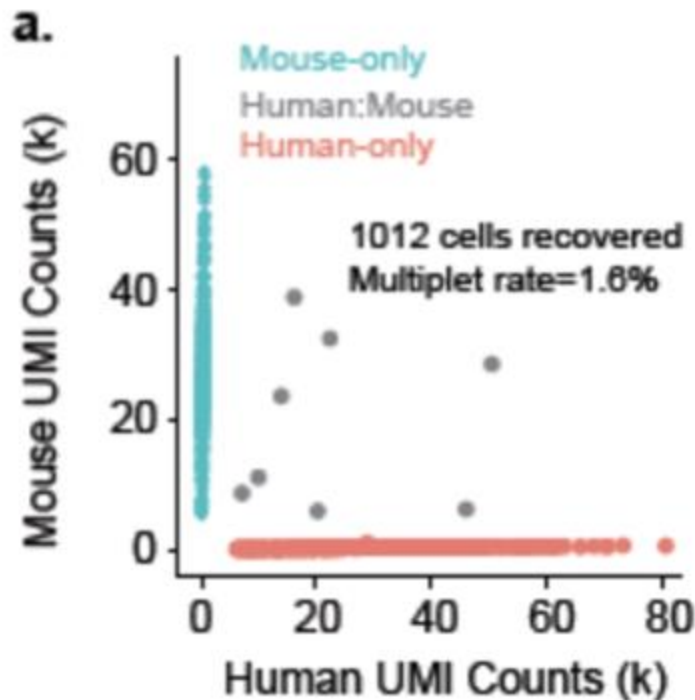


EmptyDrops builds a statistical model

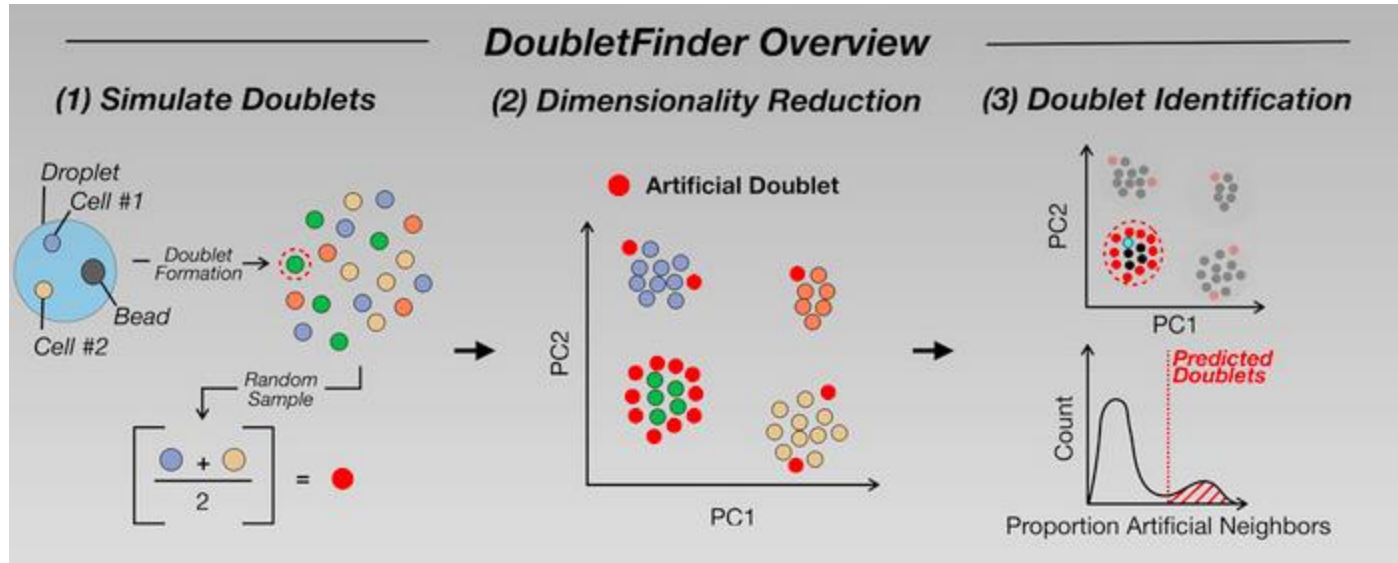


Lun et al, Genome
Biology, 2019

Doublet detection

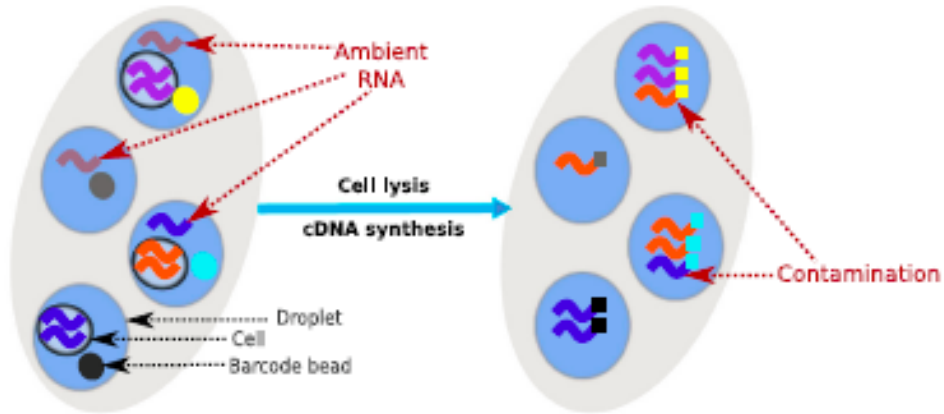


DoubletFinder and Scrublet use k-NN approach



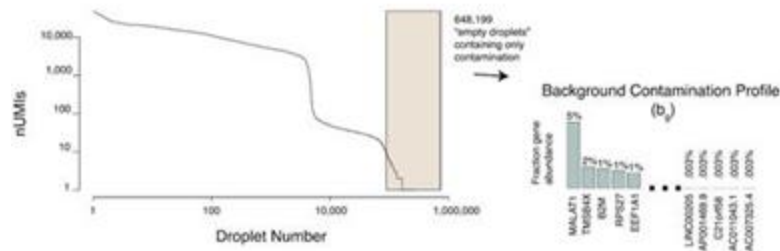
McGinnis et al,
Cell Systems, 2019

Ambient RNA



SoupX corrects for background

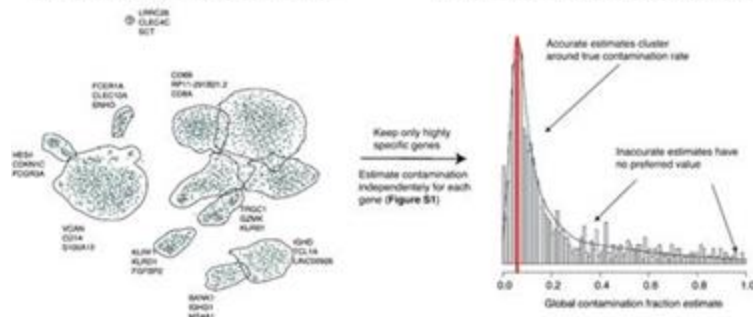
1. Determine the expression profile of contamination



2. Estimate or set the global contamination rate

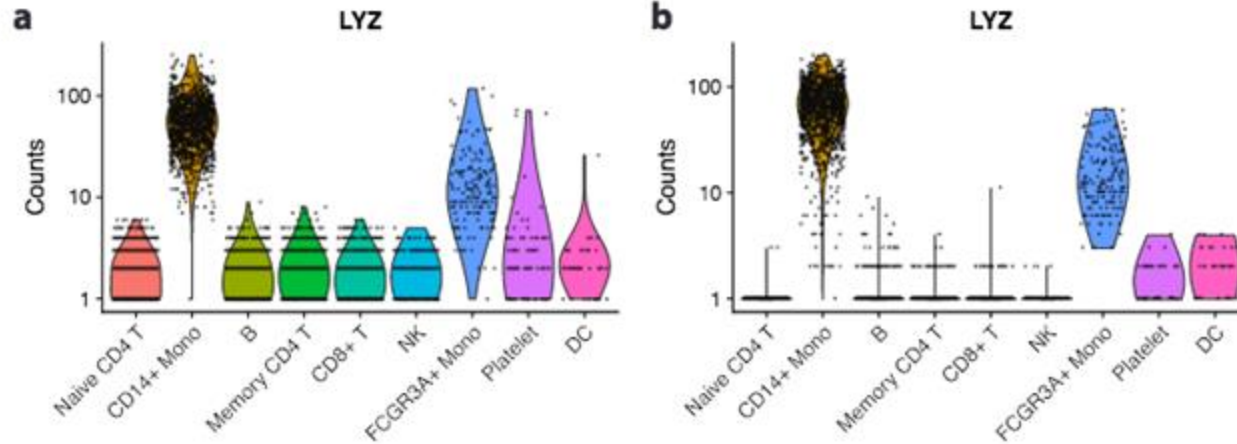
2.1 Marker genes for each cluster identified

2.2 Set contamination to most common estimate



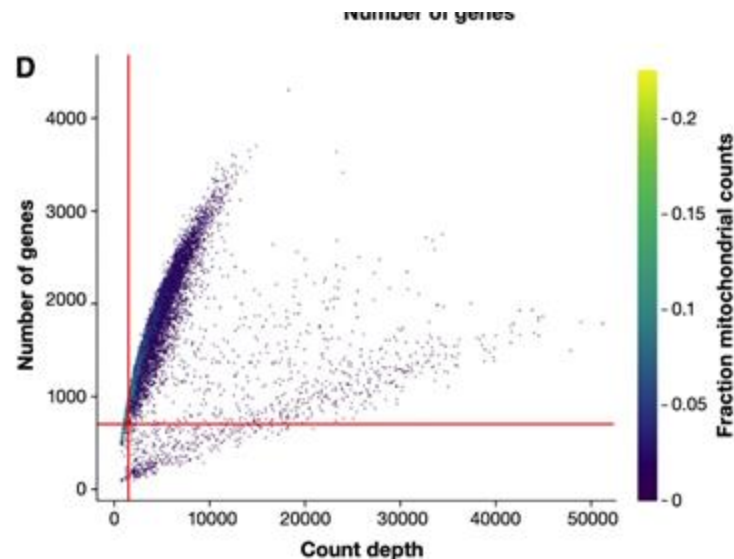
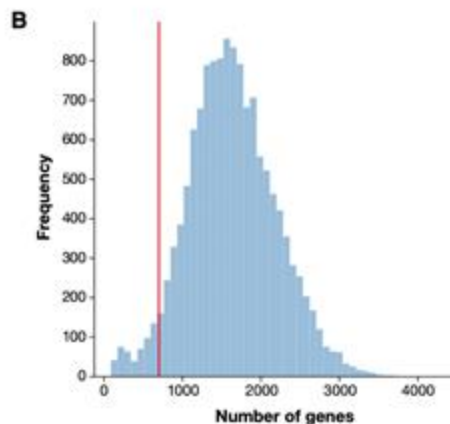
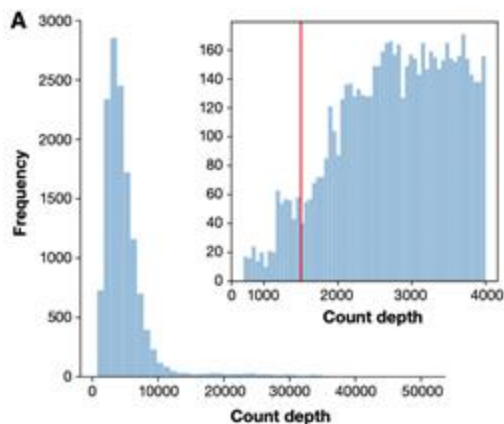
Young and Behjati,
GigaScience, 2020

CellBender uses deep autoencoders



Fleming et al,
BioRxiv, 2019

Cell QC removes low quality cells



Cleaning up the expression matrix

The expression matrix is sparse

Genes	Lman2l	2.2120425	0.00000000	0.00000000	0.67654800	0.1207469	5.0424163	32.80143146
	Lmbr1	28.5795891	0.09437713	0.00000000	0.00000000	0.00000000	3.3921710	0.31120903
	Lmbr1l	4.6895301	0.66063993	0.00000000	0.00000000	0.00000000	0.1833606	0.74690167
	Lmbrd1	16.5018371	107.21242320	0.00000000	23.19593159	13.8858954	13.5686840	25.02120578
	Lmbrd2	82.8631121	3.30319966	0.00000000	9.95492064	0.00000000	61.1507583	7.22004943
	Lmcd1	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
	Lmf1	2.2562834	8.39956485	311.6960578	0.00000000	0.00000000	0.00000000	0.00000000
	Lmf2	46.7625785	68.70655290	0.1181562	8.69847435	0.1207469	60.6923568	54.08812892
	Lmln	12.1662338	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	6.90884040
	Lmna	82.9515938	179.59968429	5.0807166	428.93143495	300.6598215	136.9703641	230.35692190
	Lmnb1	13.4934593	8.39956485	30.0116750	32.66760365	0.00000000	69.8603865	16.92977108
	Lmnb2	0.00000000	5.66262799	0.00000000	68.23469876	7.2448150	0.00000000	46.24566143
	Lmo1	76.3597071	4.90761092	0.00000000	0.00000000	0.00000000	0.00000000	3.98347555
	Lmo2	0.00000000	0.00000000	0.00000000	0.00000000	0.2414938	83.9791523	108.30074145
	Lmo3	22.9167603	0.00000000	0.00000000	0.00000000	0.1207469	0.0916803	0.06224181
	Lmo4	13.8473861	0.00000000	0.00000000	0.09664971	0.00000000	0.00000000	3.36105749
	Lmo7	641.2711210	49.07610921	0.00000000	37.11349054	0.2414938	0.00000000	9.58523804
	Lmod1	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
	Lmod3	0.9732987	3.68070819	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
	Lmtk2	27.6947721	52.00180034	218.5889716	12.75776237	0.00000000	62.7093233	60.43679307
	Lmtk3	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
Cells	Lmx1a	0.0884817	1.41565700	0.00000000	0.00000000	0.00000000	0.00000000	0.06224181
	Lnp	19.7756600	114.29070818	0.5907810	17.88019727	32.9639081	74.8111225	5.16606985
	Lnppep	15.9267060	5.19074232	25.2854270	0.00000000	0.00000000	1.2835242	5.41503707
	Lnx1	28.4911074	0.28313140	0.00000000	0.00000000	0.00000000	0.00000000	0.18672542
	Lnx2	112.1505548	73.51978669	0.00000000	0.00000000	0.1207469	0.4584015	1.12035250
	Loh12cr1	134.3594615	93.15023037	0.00000000	0.00000000	0.00000000	0.00000000	0.06224181
	Lonp1	114.3625973	35.86331058	111.0668288	5.21908461	2.4149383	43.5481412	326.76947851
	Lonp2	85.6502856	57.66442832	0.00000000	14.69075667	0.1207469	5.1340966	124.23464364
	Lonrf1	38.0028902	0.00000000	0.00000000	0.00000000	0.00000000	54.5497769	3.54778291

Some zeros represent absent genes

Lman2l	2.2120425	0.00000000	0.00000000	0.67654800	0.1207469	5.0424163	32.80143146
Lmbr1	28.5795891	0.09437713	0.00000000	0.00000000	0.00000000	3.3921710	0.31120903
Lmbr1l	4.6895301	0.66063993	0.00000000	0.00000000	0.00000000	0.1833606	0.74690167
Lmbrd1	16.5018371	107.21242320	0.00000000	23.19593159	13.8858954	13.5686840	25.02120578
Lmbrd2	82.8631121	3.30319966	0.00000000	9.95492064	0.00000000	61.1507583	7.22004943
Lmcd1	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
Lmf1	2.2562834	8.39956485	311.6960578	0.00000000	0.00000000	0.00000000	0.00000000
Lmf2	46.7625785	68.70655290	0.1181562	8.69847435	0.1207469	60.6923568	54.08812892
Lmln	12.1662338	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	6.90884040
Lmna	82.9515938	179.59968429	5.0807166	428.93143495	300.6598215	136.9703641	230.35692190
Lmnb1	13.4934593	8.39956485	30.0116750	32.66760365	0.00000000	69.8603865	16.92977108
Lmnb2	0.00000000	5.66262799	0.00000000	68.23469876	7.2448150	0.00000000	46.24566143
Lmo1	76.3597071	4.90761092	0.00000000	0.00000000	0.00000000	0.00000000	3.98347555
Lmo2	0.00000000	0.00000000	0.00000000	0.00000000	0.2414938	83.9791523	108.30074145
Lmo3	22.9167603	0.00000000	0.00000000	0.00000000	0.1207469	0.0916803	0.06224181
Lmo4	13.8473861	0.00000000	0.00000000	0.09664971	0.00000000	0.00000000	3.36105749
Lmo7	641.2711210	49.07610921	0.00000000	37.11349054	0.2414938	0.00000000	9.58523804
Lmod1	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
Lmod3	0.9732987	3.68070819	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
Lmtk2	27.6947721	52.00180034	218.5880716	12.75776237	0.00000000	62.7093233	60.43679307
Lmtk3	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
Lmx1a	0.0884817	1.41565700	0.00000000	0.00000000	0.00000000	0.00000000	0.06224181
Lnp	19.7756600	114.29070818	0.5907810	17.88019727	32.9639081	74.8111225	5.16606985
Lnpep	15.9267060	5.19074232	25.2854270	0.00000000	0.00000000	1.2835242	5.41503707
Lnx1	28.4911074	0.28313140	0.00000000	0.00000000	0.00000000	0.00000000	0.18672542
LnX2	112.1505548	73.51978669	0.00000000	0.00000000	0.1207469	0.4584015	1.12035250
Loh12cr1	134.3594615	93.15023037	0.00000000	0.00000000	0.00000000	0.00000000	0.06224181
Lonp1	114.3625973	35.86331058	111.0668288	5.21908461	2.4149383	43.5481412	326.76947851
Lonp2	85.6502856	57.66442832	0.00000000	14.69075667	0.1207469	5.1340966	124.23464364
Lonrf1	38.0028902	0.00000000	0.00000000	0.00000000	0.00000000	54.5497769	3.54778291

Genes

Cells

Some zeros represent missing data

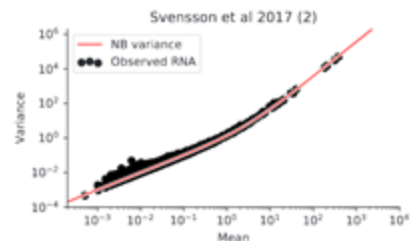
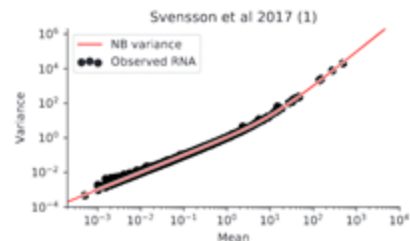
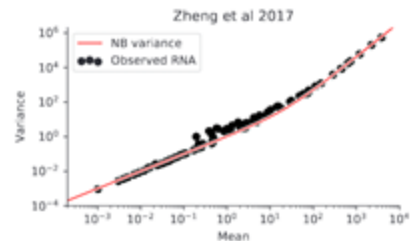
Lman2l	2.2120425	0.00000000	0.00000000	0.67654800	0.1207469	5.0424163	32.80143146
Lmbr1	28.5795891	0.09437713	0.00000000	0.00000000	0.00000000	3.3921710	0.31120903
Lmbr1l	4.6895301	0.66063993	0.00000000	0.00000000	0.00000000	0.1833606	0.74690167
Lmbrd1	16.5018371	107.21242320	0.00000000	23.19593159	13.8858954	13.5686840	25.02120578
Lmbrd2	82.8631121	3.30319966	0.00000000	9.95492064	0.00000000	61.1507583	7.22004943
Lmcd1	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
Lmf1	2.2562834	8.39956485	311.6960578	0.00000000	0.00000000	0.00000000	0.00000000
Lmf2	46.7625785	68.70655290	0.1181562	8.69847435	0.1207469	60.6923568	54.08812892
Lmln	12.1662338	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	6.90884040
Lmna	82.9515938	179.59968429	5.0807166	428.93143495	300.6598215	136.9703641	230.35692190
Lmnb1	13.4934593	8.39956485	30.0116750	32.66760365	0.00000000	60.8605865	16.92977108
Lmnb2	0.00000000	5.66262799	0.00000000	68.23469876	7.2448150	0.00000000	16.24566143
Lmo1	76.3597071	4.90761092	0.00000000	0.00000000	0.00000000	0.00000000	3.98347555
Lmo2	0.00000000	0.00000000	0.00000000	0.00000000	0.2414938	83.9791523	108.30074145
Lmo3	22.9167603	0.00000000	0.00000000	0.00000000	0.1207469	0.0916803	0.06224181
Lmo4	13.8473861	0.00000000	0.00000000	0.09664971	0.00000000	0.00000000	3.36105749
Lmo7	641.2711210	49.07610921	0.00000000	37.11349054	0.2414938	0.00000000	9.58523804
Lmod1	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
Lmod3	0.9732987	3.68070819	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
Lmtk2	27.6947721	52.00180034	218.5880716	12.75776237	0.00000000	62.7093233	60.43679307
Lmtk3	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
Lmx1a	0.0884817	1.41565700	0.00000000	0.00000000	0.00000000	0.00000000	0.06224181
Lnp	19.7756600	114.29070818	0.5907810	17.88019727	32.9639081	74.8111225	5.16606985
Lnpep	15.9267060	5.19074232	25.2854270	0.00000000	0.00000000	1.2835242	5.41503707
LnX1	28.4911074	0.28313140	0.00000000	0.00000000	0.00000000	0.00000000	0.18672542
LnX2	112.1505548	73.51978669	0.00000000	0.00000000	0.1207469	0.4584015	1.12035250
Loh12cr1	134.3594615	93.15023037	0.00000000	0.00000000	0.00000000	0.00000000	0.06224181
Lonp1	114.3625973	35.86331058	111.0668288	5.21908461	2.4149383	43.5481412	326.76947851
Lonp2	85.6502856	57.66442832	0.00000000	14.69075667	0.1207469	5.1340966	124.23464364
Lonrf1	38.0028902	0.00000000	0.00000000	0.00000000	0.00000000	54.5497769	3.54778291

Genes

Cells

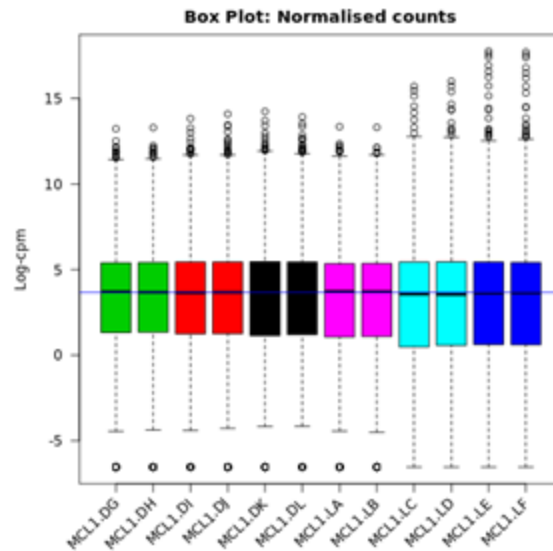
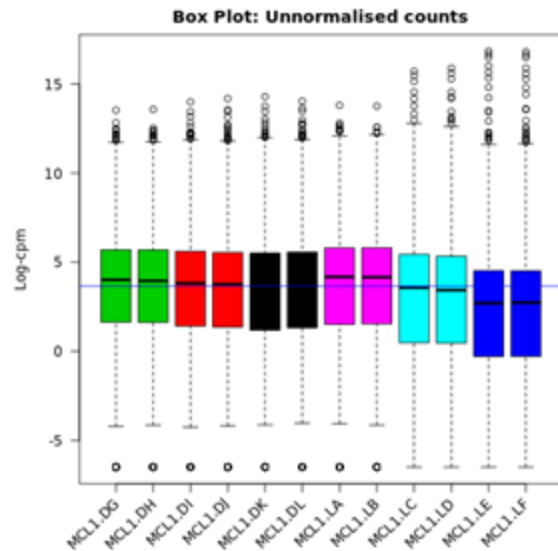
Genes follow negative binomial

$$\begin{aligned}
 f(k; r, p) &= \int_0^{\infty} f_{\text{Poisson}(\lambda)}(k) \cdot f_{\text{Gamma}\left(r, \frac{1-p}{p}\right)}(\lambda) d\lambda \\
 &= \int_0^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} \cdot \lambda^{r-1} \frac{e^{-\lambda(1-p)/p}}{\left(\frac{p}{1-p}\right)^r \Gamma(r)} d\lambda \\
 &= \frac{(1-p)^r p^{-r}}{k! \Gamma(r)} \int_0^{\infty} \lambda^{r+k-1} e^{-\lambda/p} d\lambda \\
 &= \frac{(1-p)^r p^{-r}}{k! \Gamma(r)} p^{r+k} \Gamma(r+k) \\
 &= \frac{\Gamma(r+k)}{k! \Gamma(r)} p^k (1-p)^r.
 \end{aligned}$$



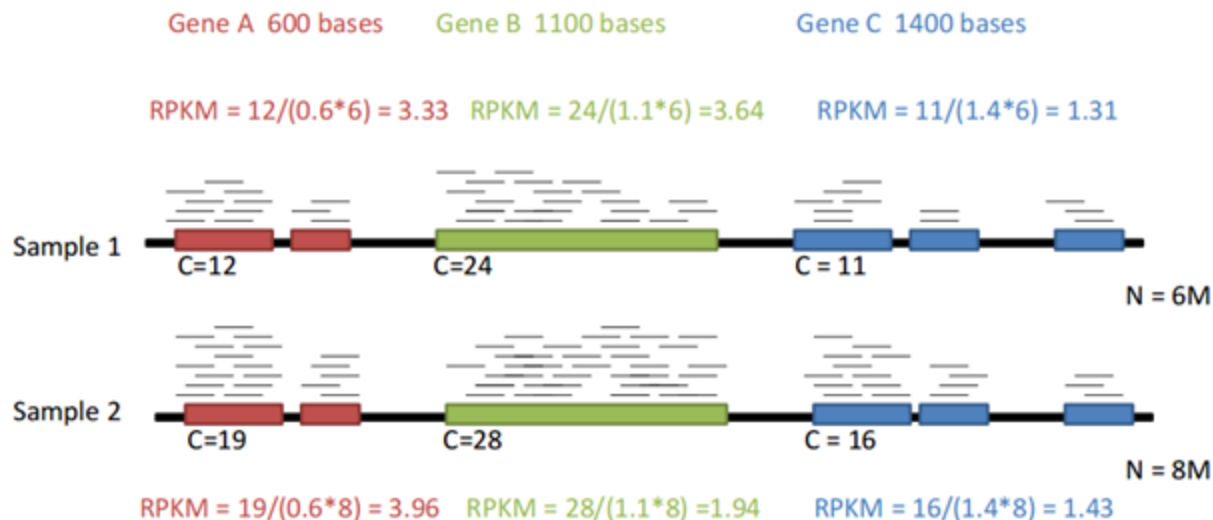
Svensson, Nat
Biotech, 2020

Normalization

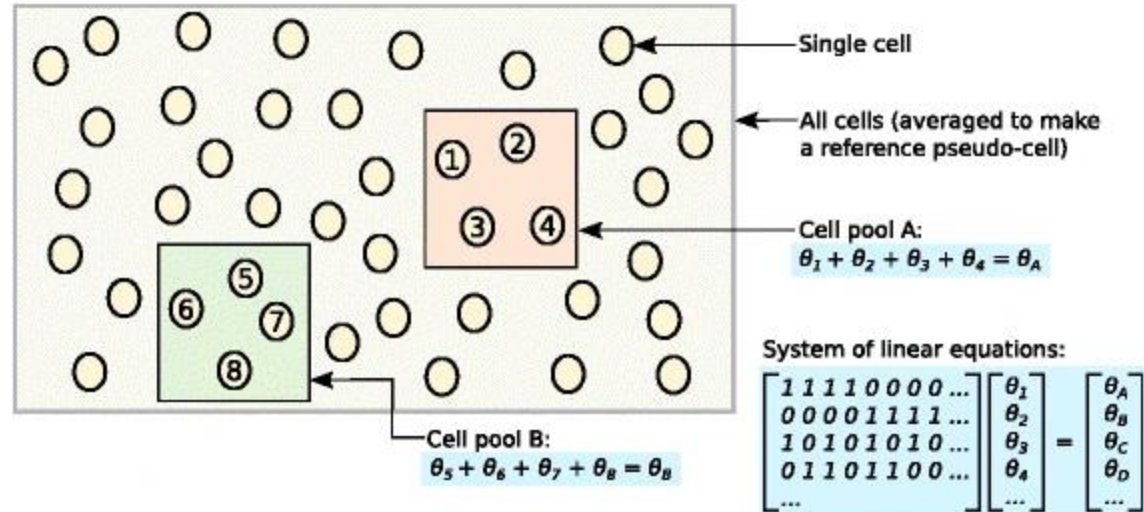


Methods from bulk RNA-seq

RPKM Example

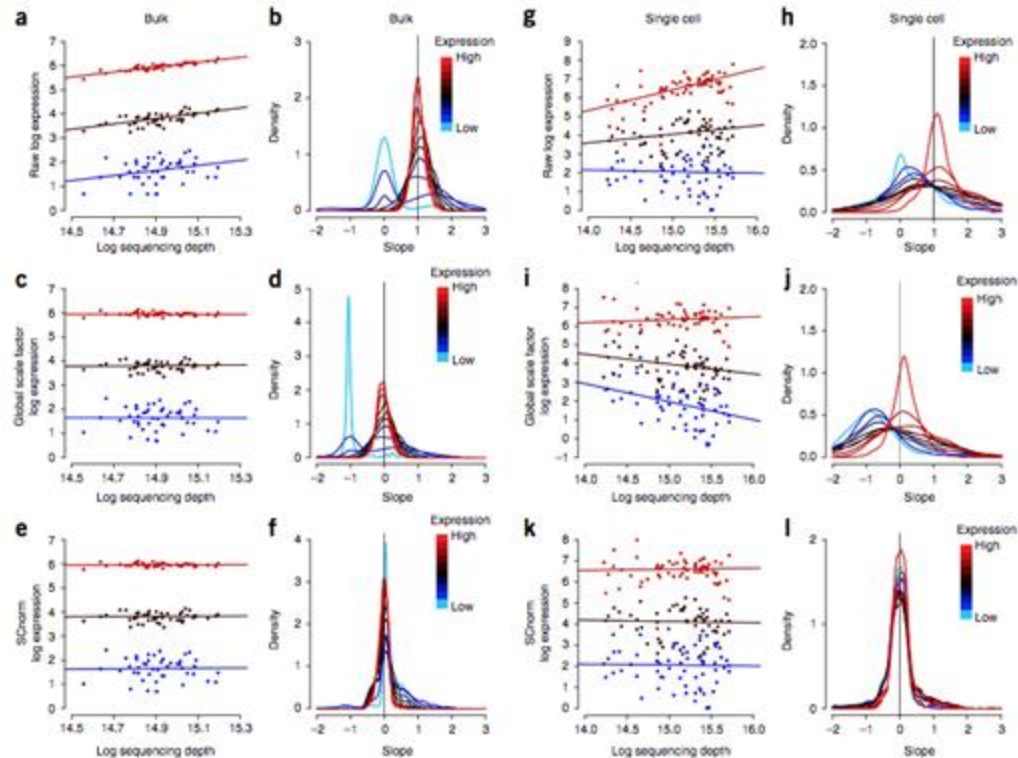


Scran uses pools of cells



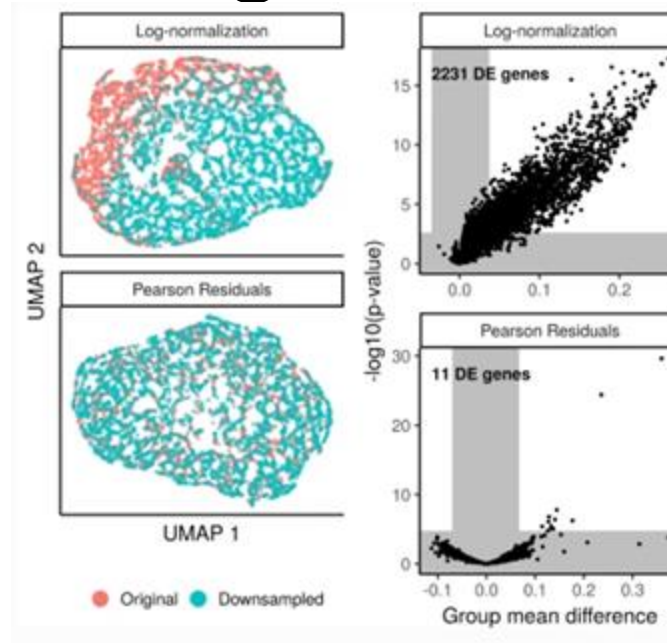
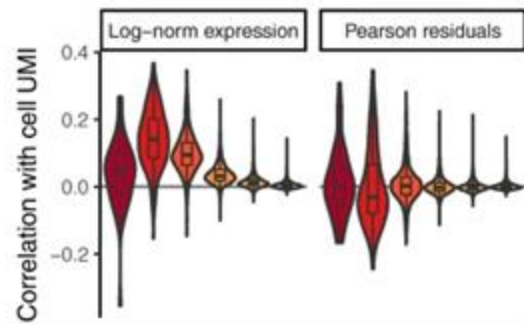
Lun et al, Genome
Biology, 2016

SCnorm regresses sequencing depth



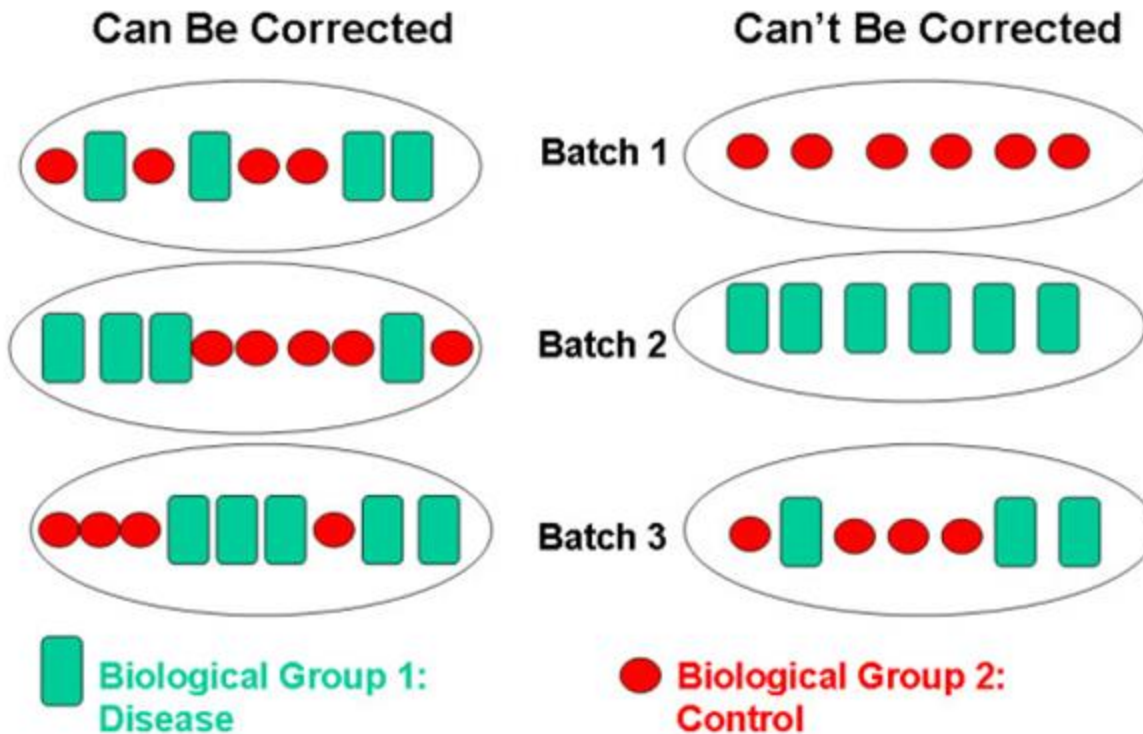
Bacher et al, Nat
Methods, 2017

Sctransform uses residuals from negative binomial regression

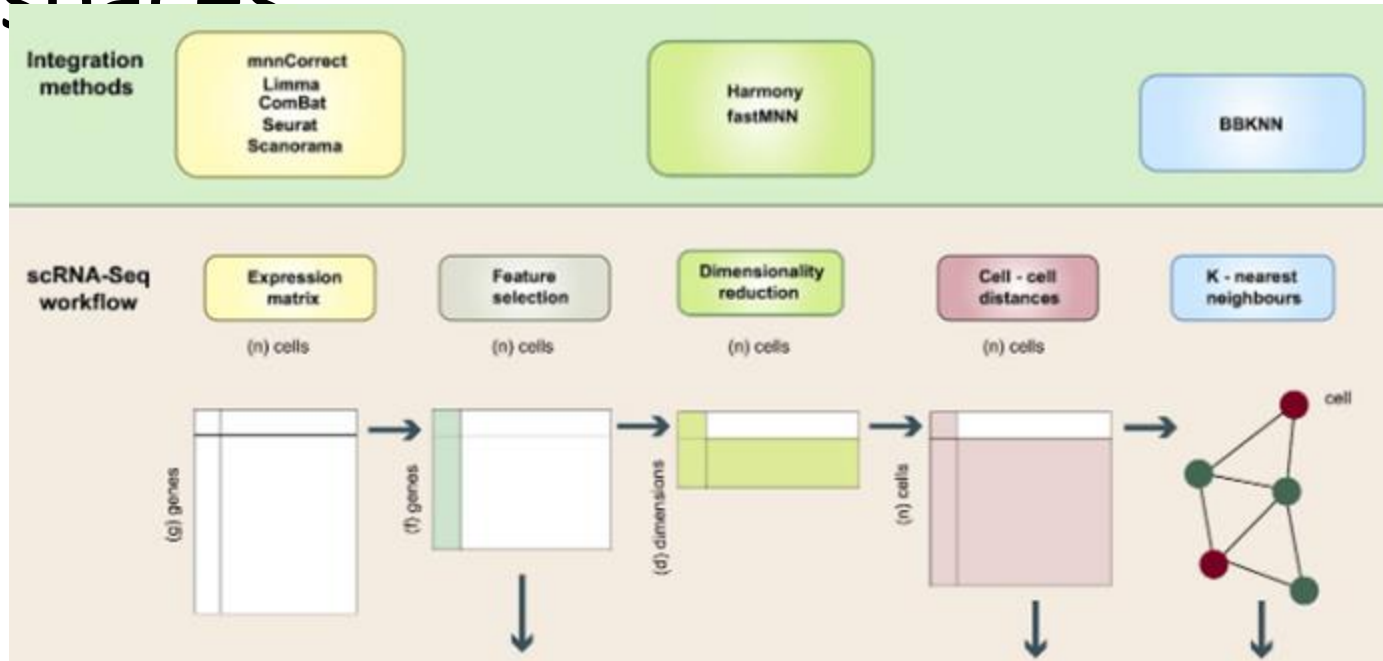


Hafemeister and
Satija, Genome
Biology, 2019

Batch correction

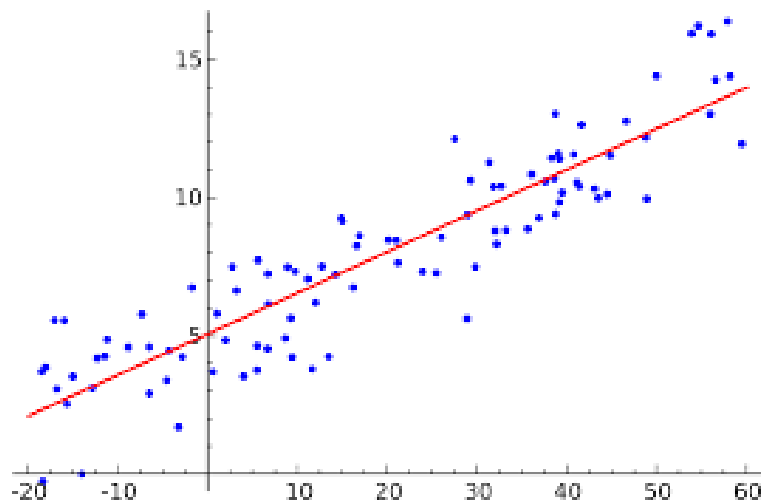


Corrections can be in different spaces



Chazarra-Gil et al,
NAR, 2021

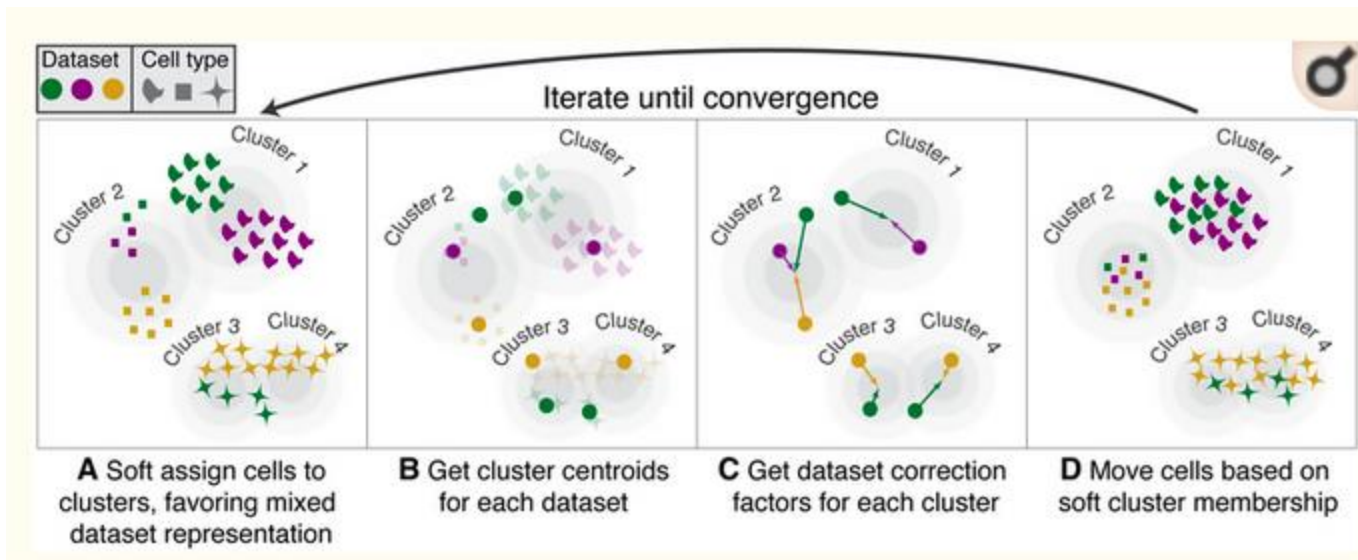
Combat regresses out batch



Sorry! Could not come up with a more interesting figure

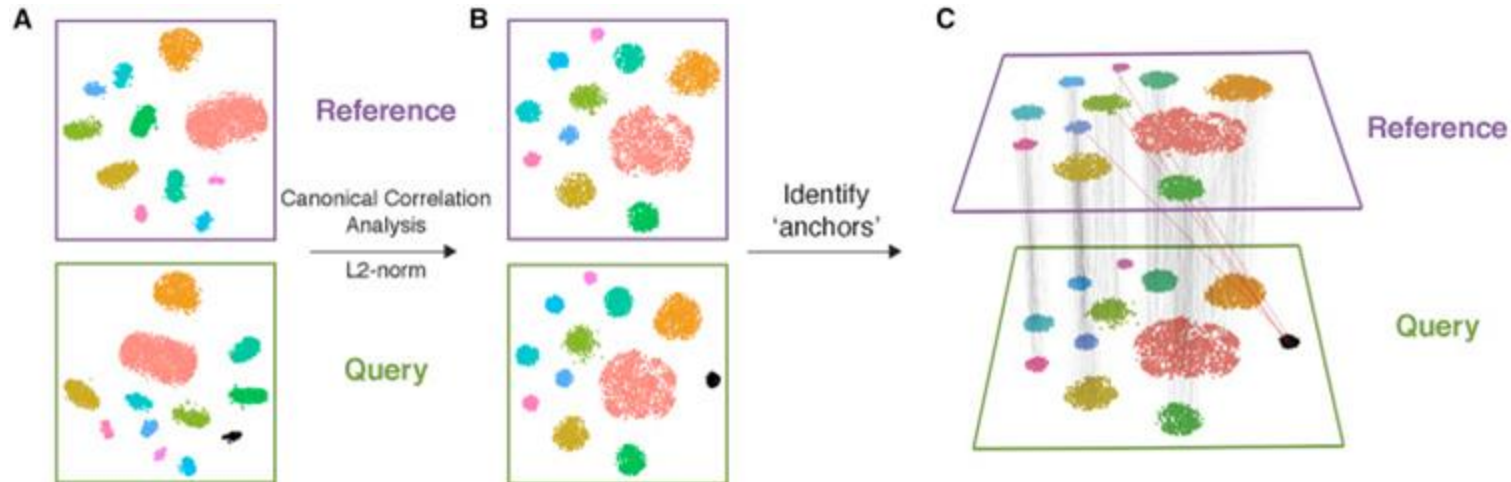
Johnson et al,
Biostatistics, 2007

Harmony



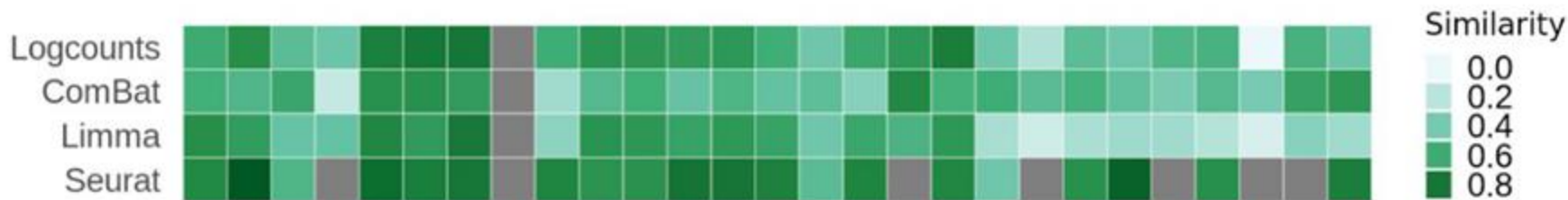
Korsunsky et al, Nat
Methods, 2019

Seurat anchor method



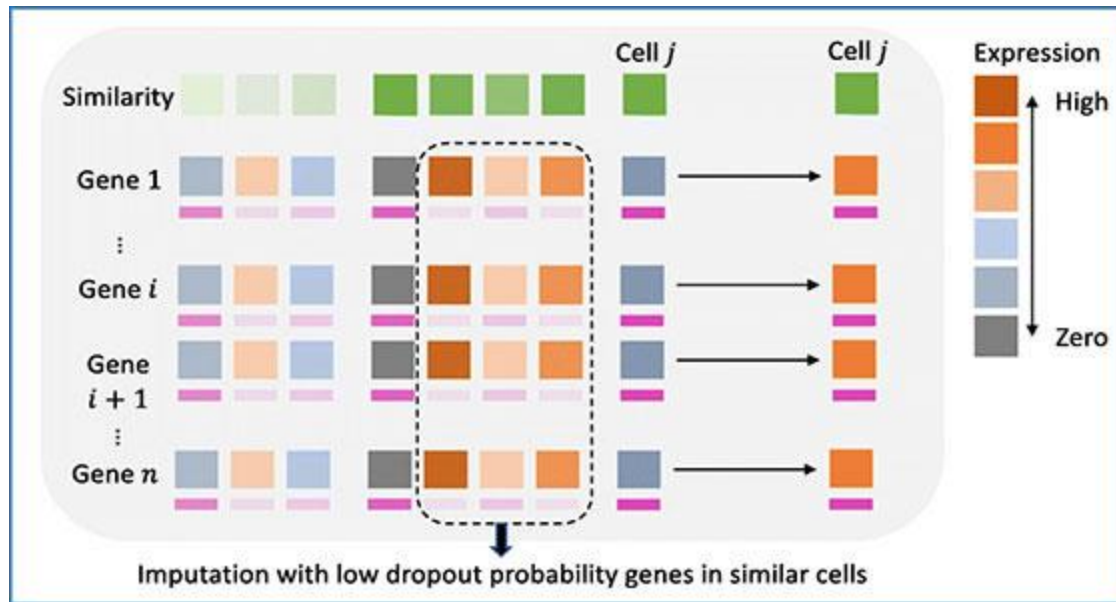
Butler et al, Cell,
2019

Correction can distort expression

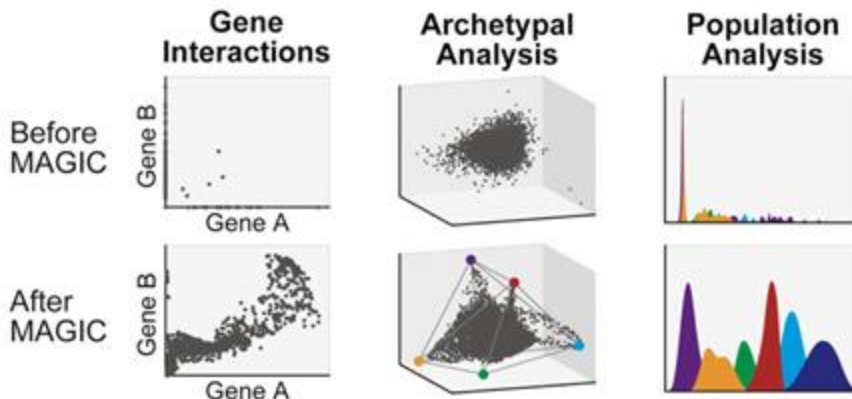
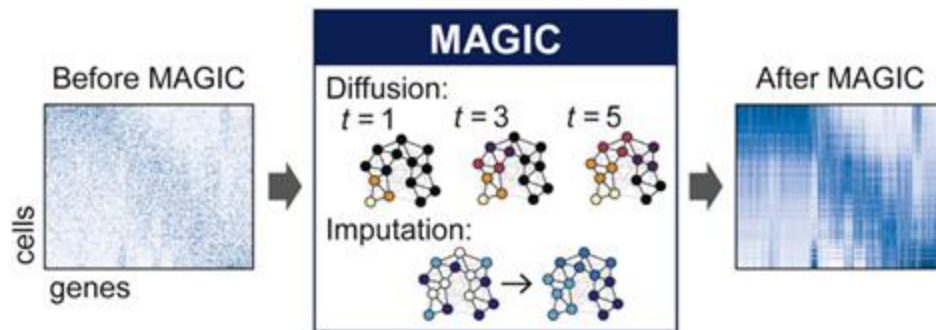


Chazarra-Gil et al,
NAR, 2021

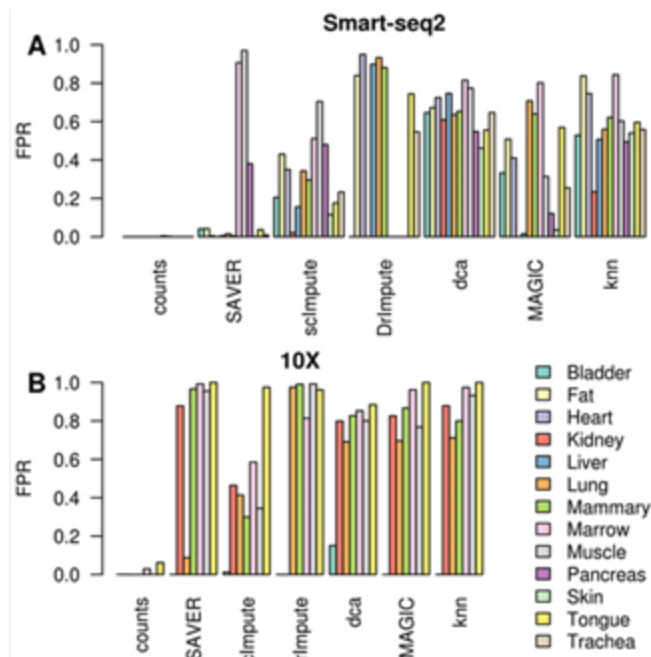
Imputation of dropouts



MAGIC uses Markov diffusion

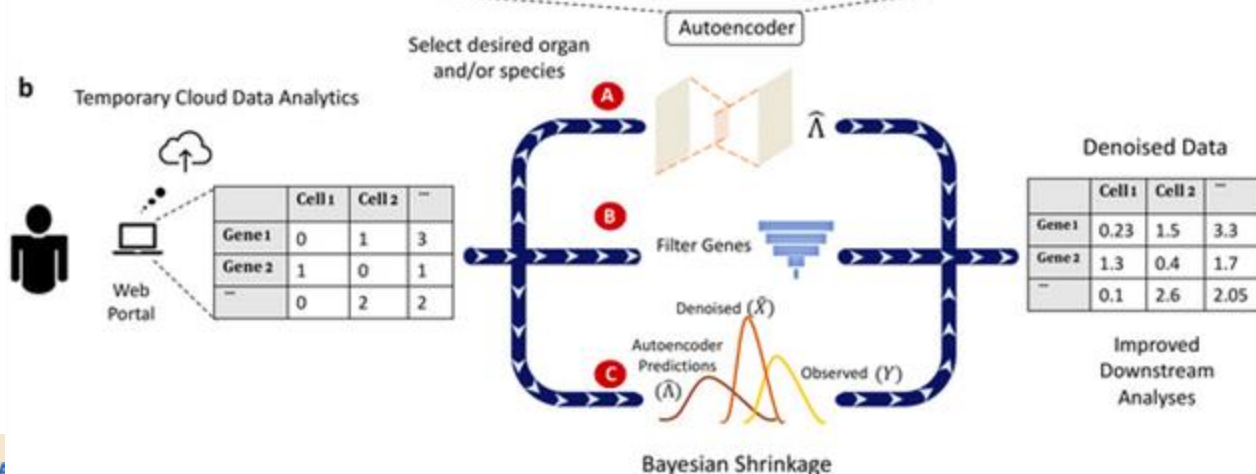
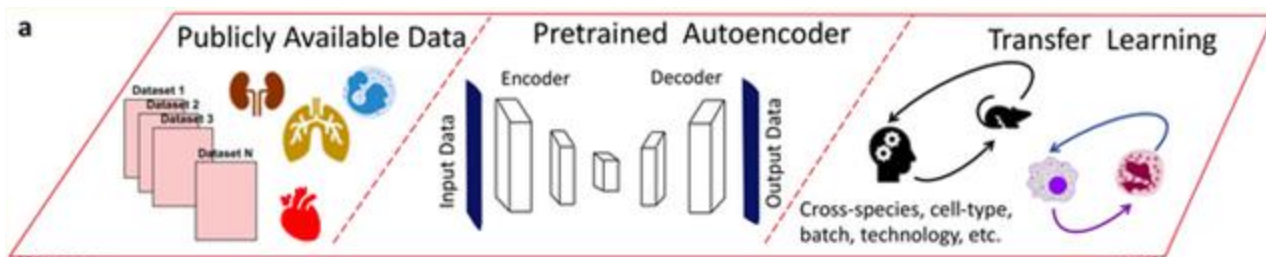


Imputation creates false positives

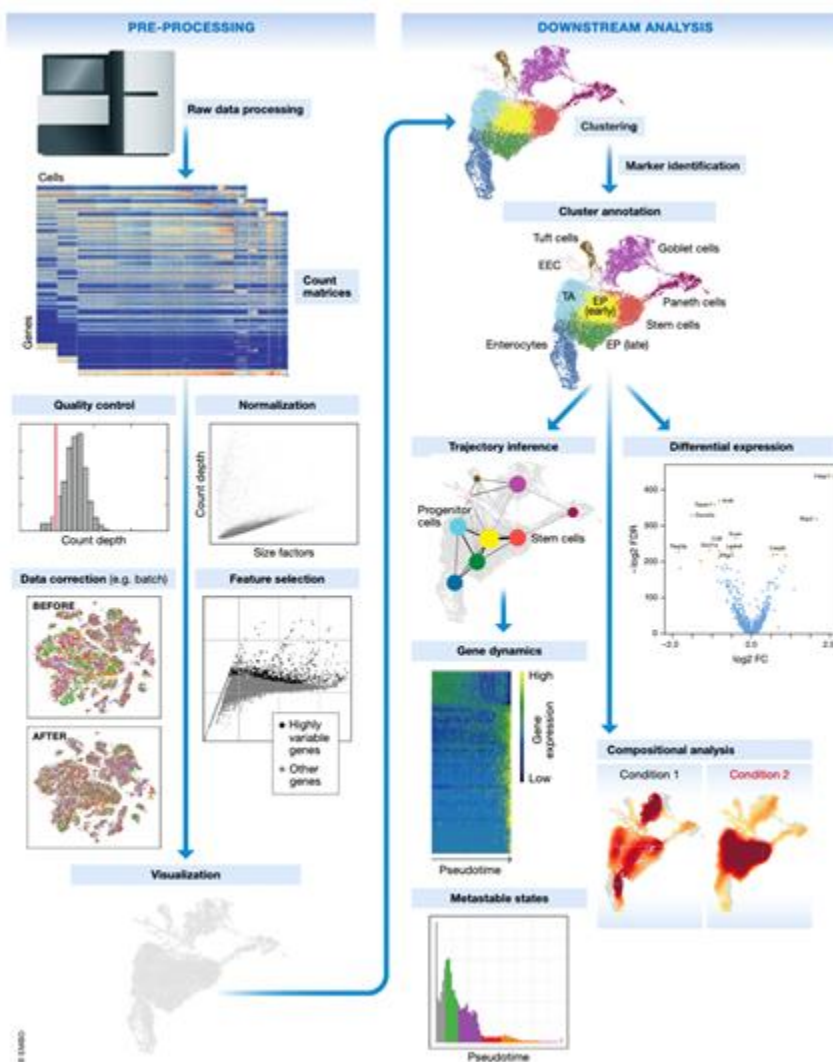


Andrews and Hemberg,
F1000 research, 2019

Impute from external reference



Wang et al, Nat
Methods, 2019

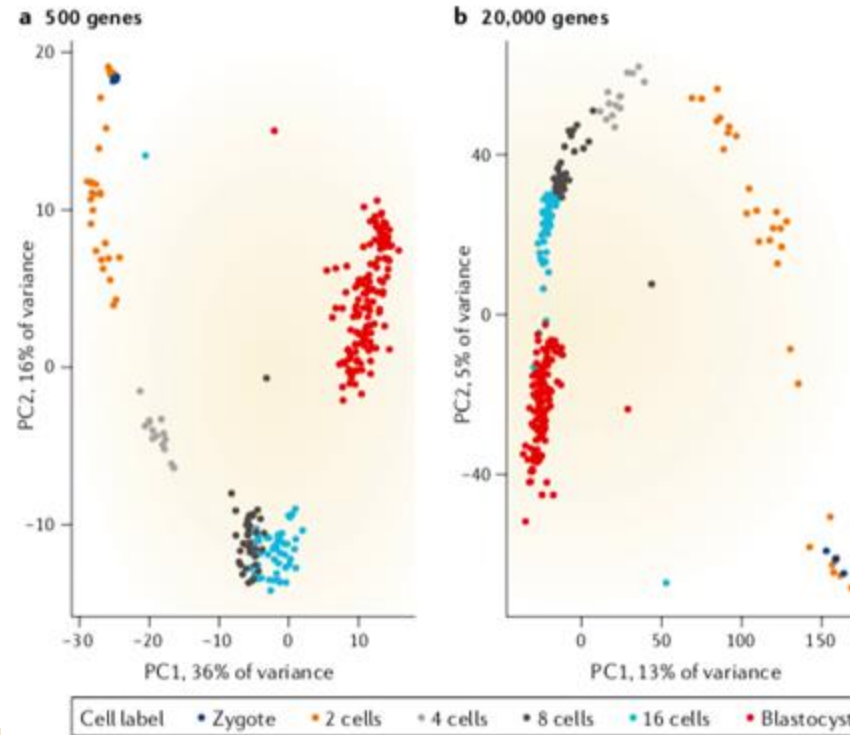


Luecken and Theis,
MSB, 2019

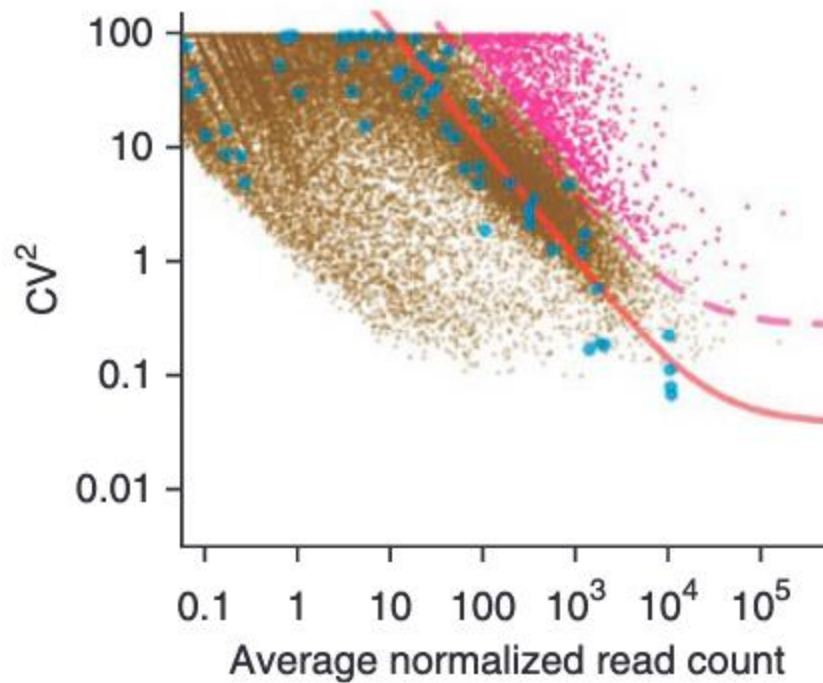
Single cell RNA-seq, part II

Martin Hemberg
STAT115/215, BIO/BST282
March 25, 2021

The curse of dimensionality

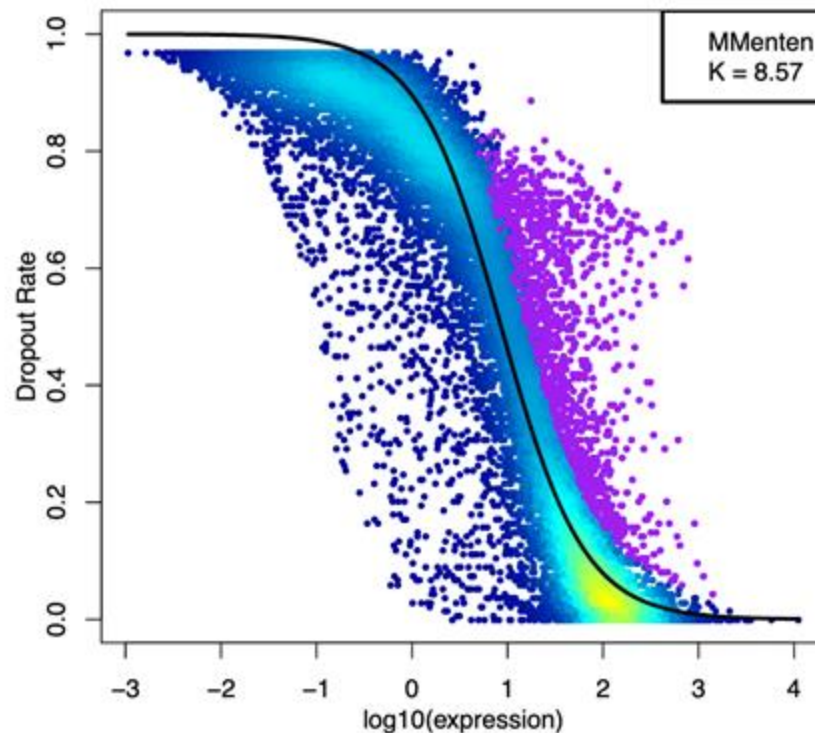


Select highly variable genes



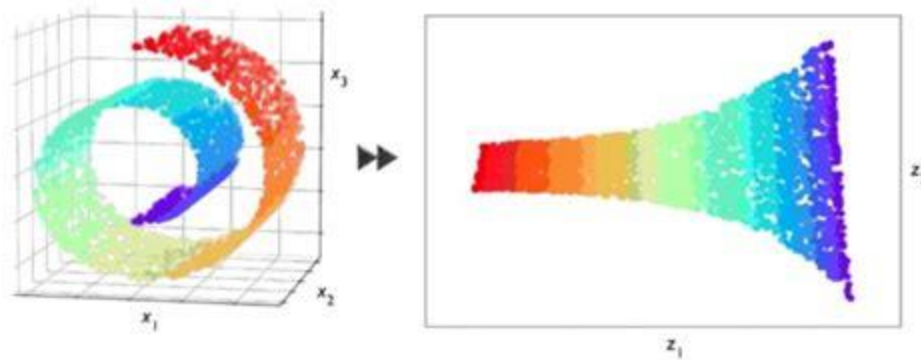
Brennecke et al, Nat
Methods, 2013

Select high dropout genes



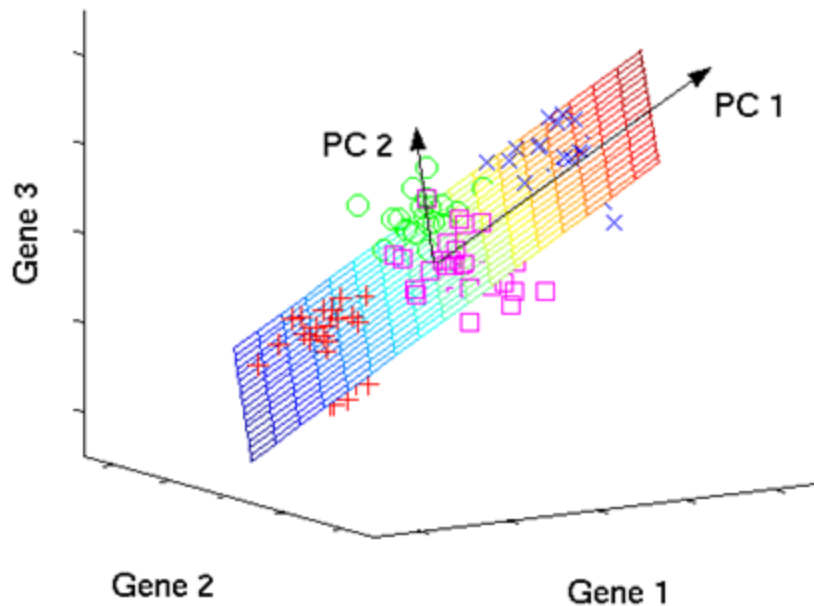
Andrews and Hemberg,
Bioinformatics, 2018

Dimensionality reduction



Principal component analysis

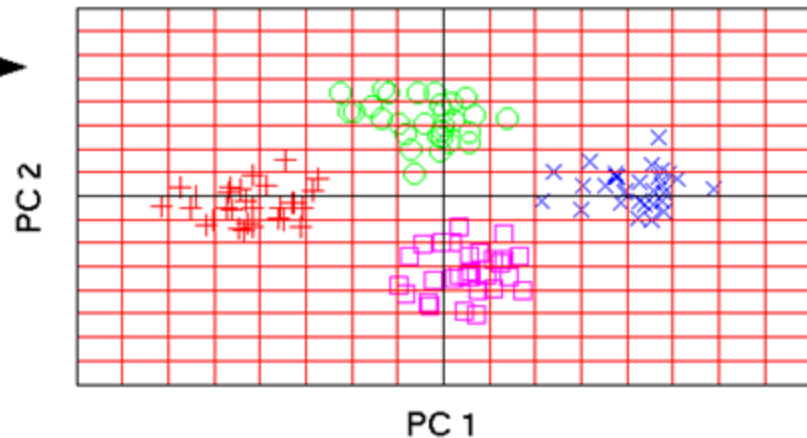
original data space



PCA



component space

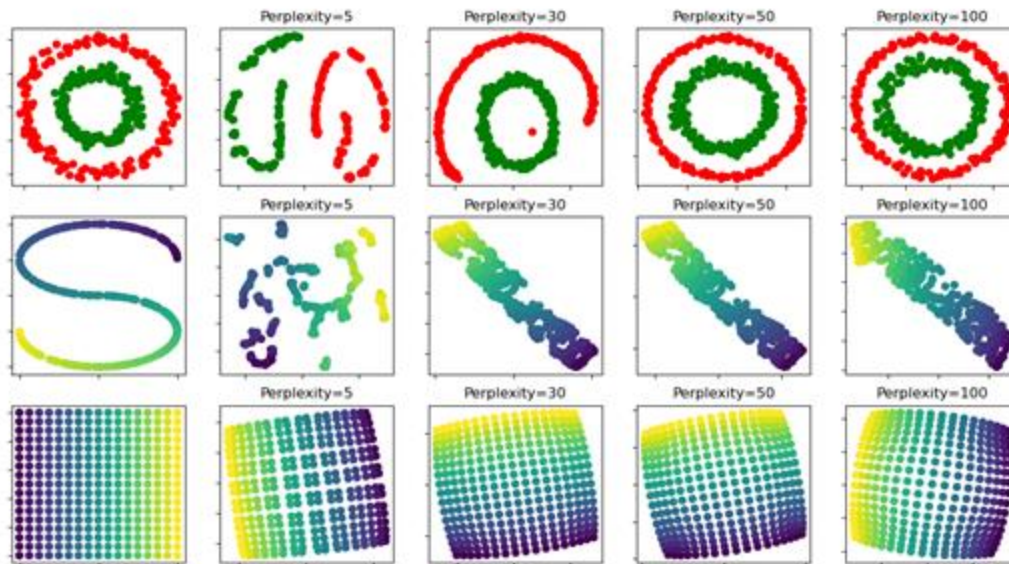


Visualization



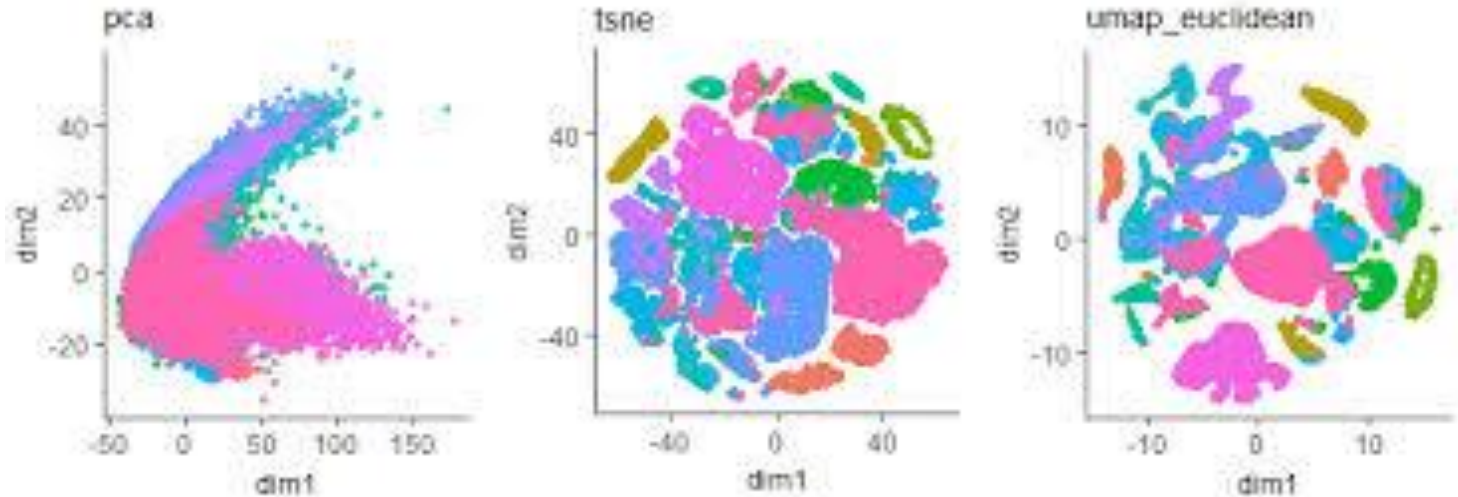
This is not easy to illustrate either, so please bear with me

tSNE uses a non-linear projection



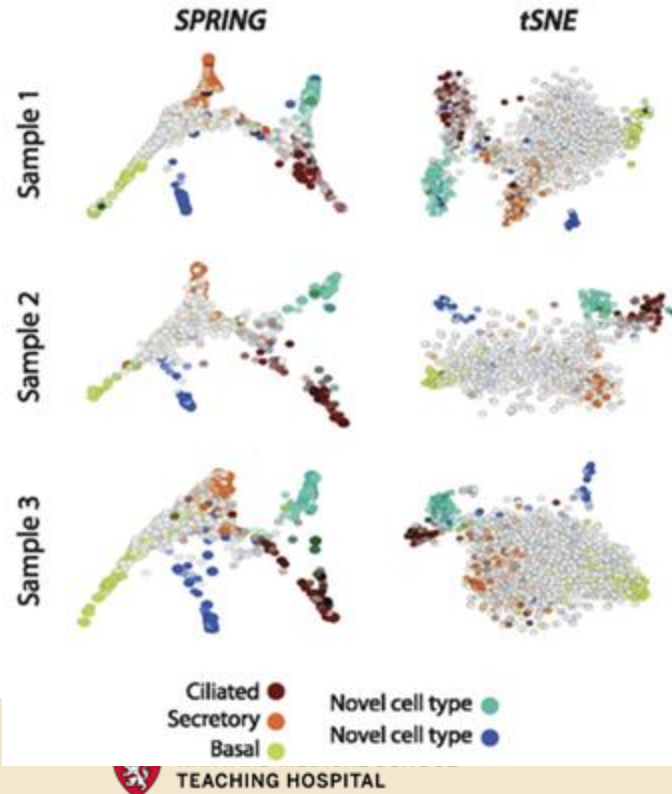
van der Maaten and
Hinton, JMLR, 2008

UMAP better preserves global structure



Becht et al, Nat
Biotech, 2019

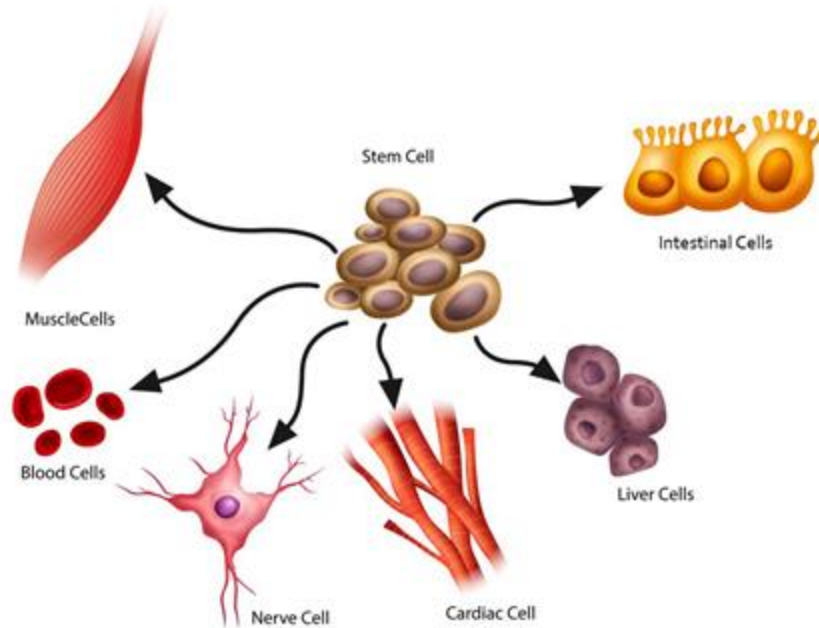
Force directed graph layout



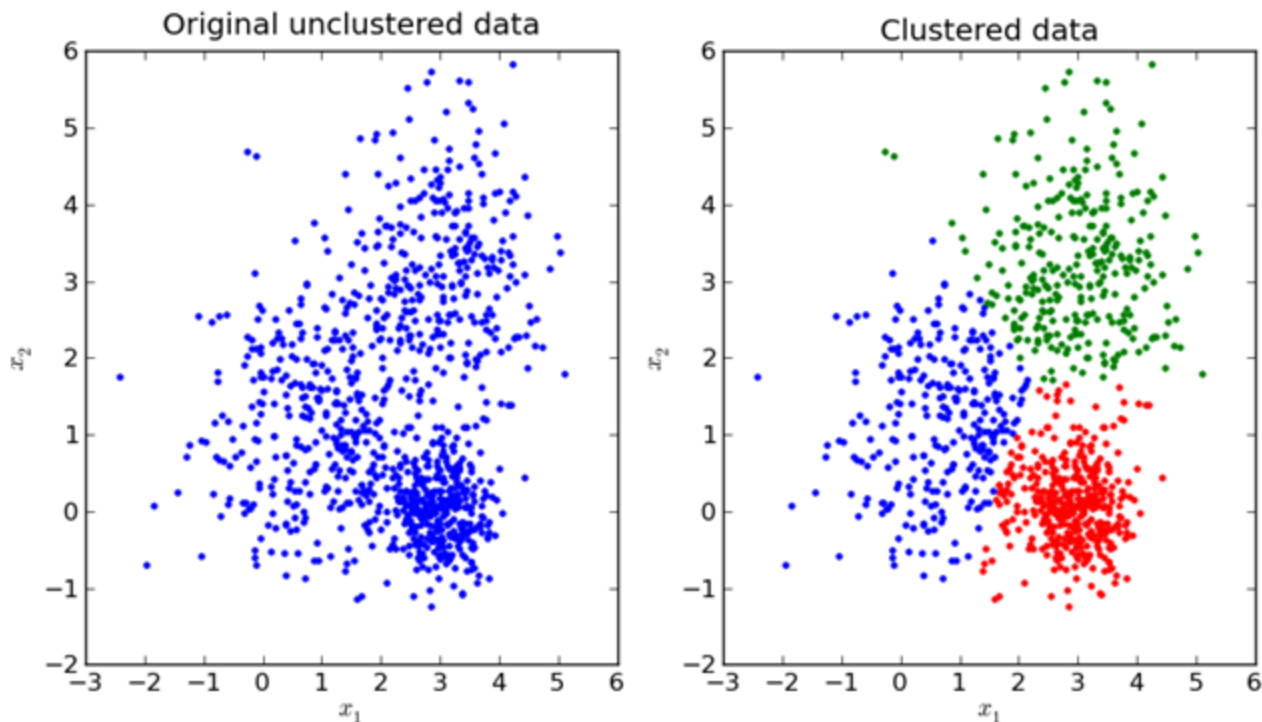
Weinreb et al,
Bioinformatics, 2018

Clustering and projections

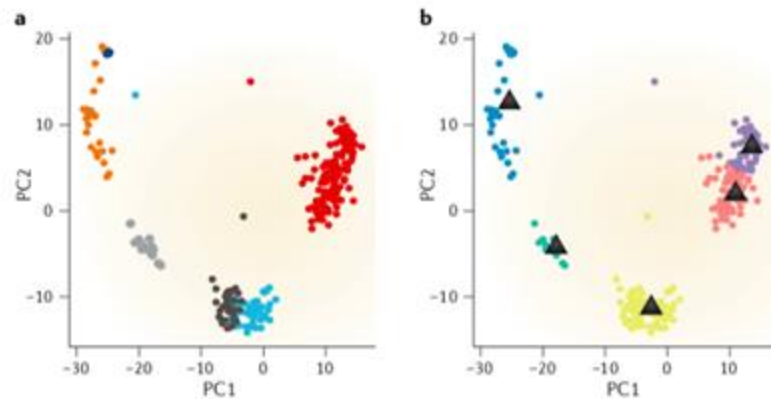
Definition of cell-type based on transcriptome



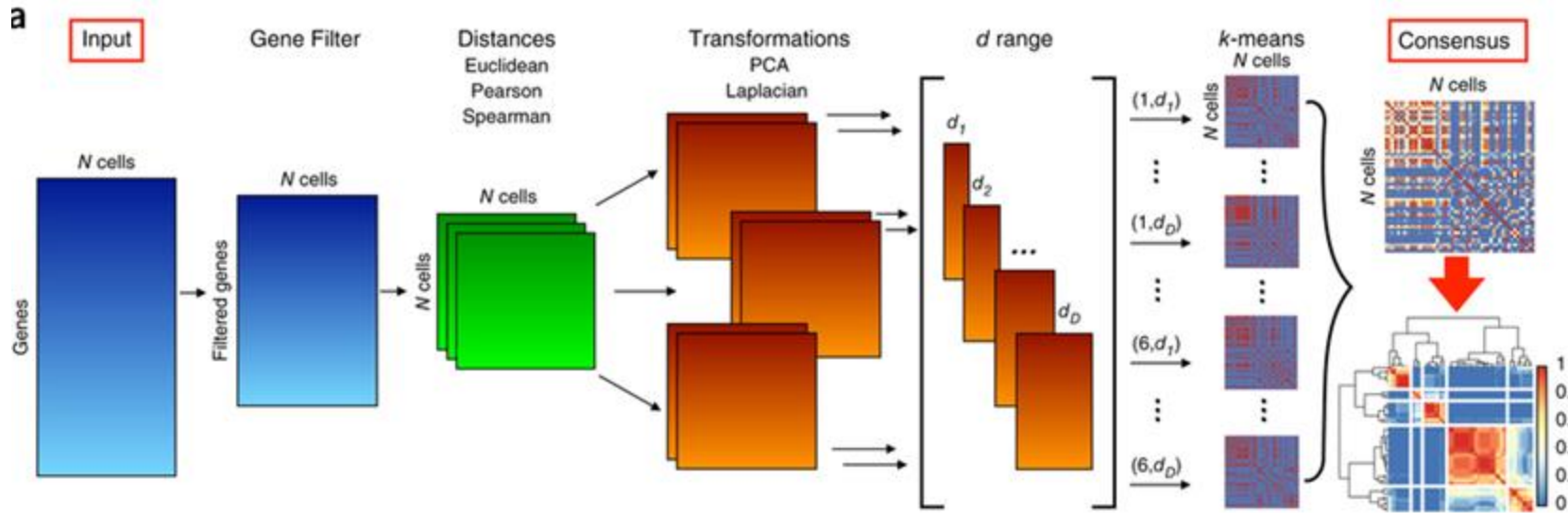
Unsupervised clustering



K-means based strategies

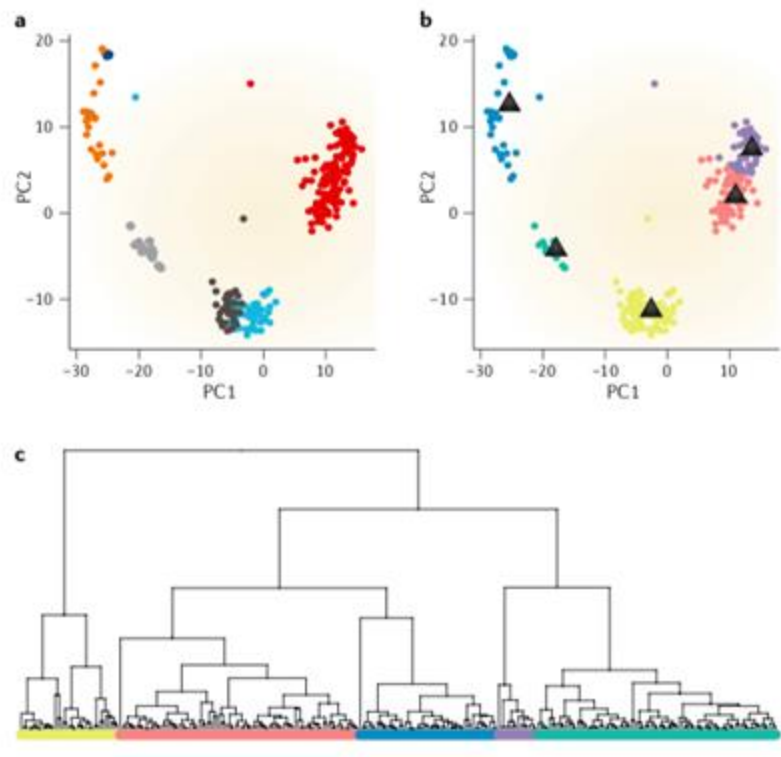


Single-cell consensus clustering (SC3)



Kiselev et al, Nat
Methods, 2017

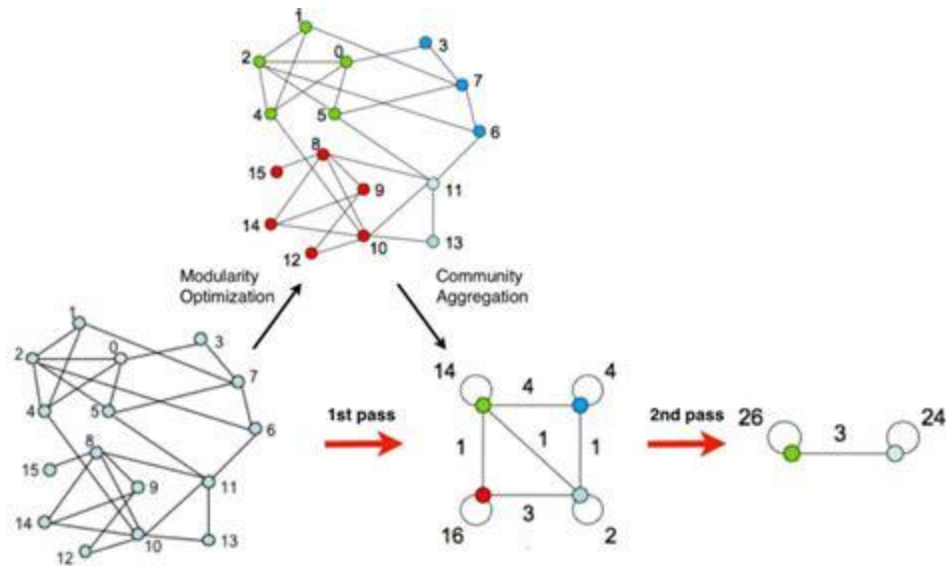
Hierarchical clustering



Graph clustering

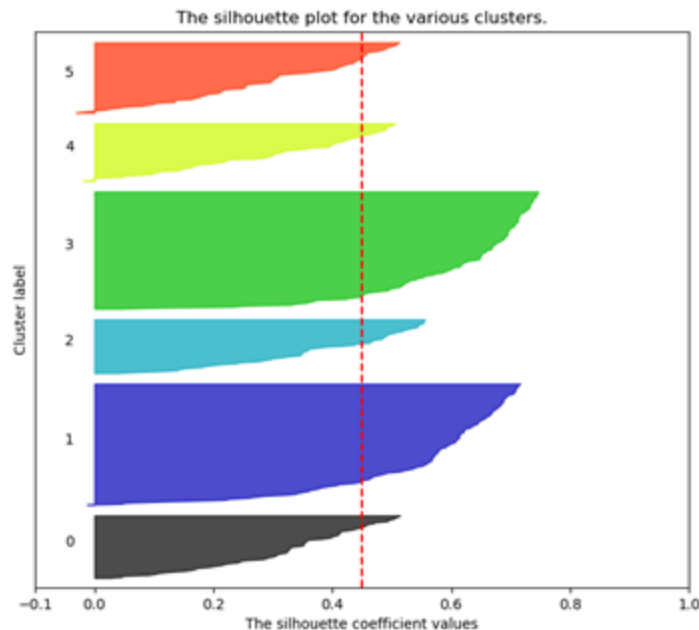


Louvain clustering

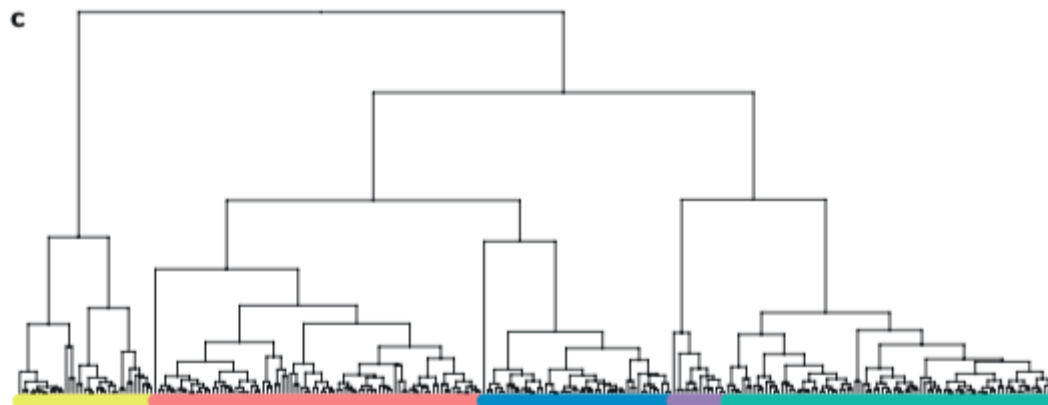


Evaluate cluster quality

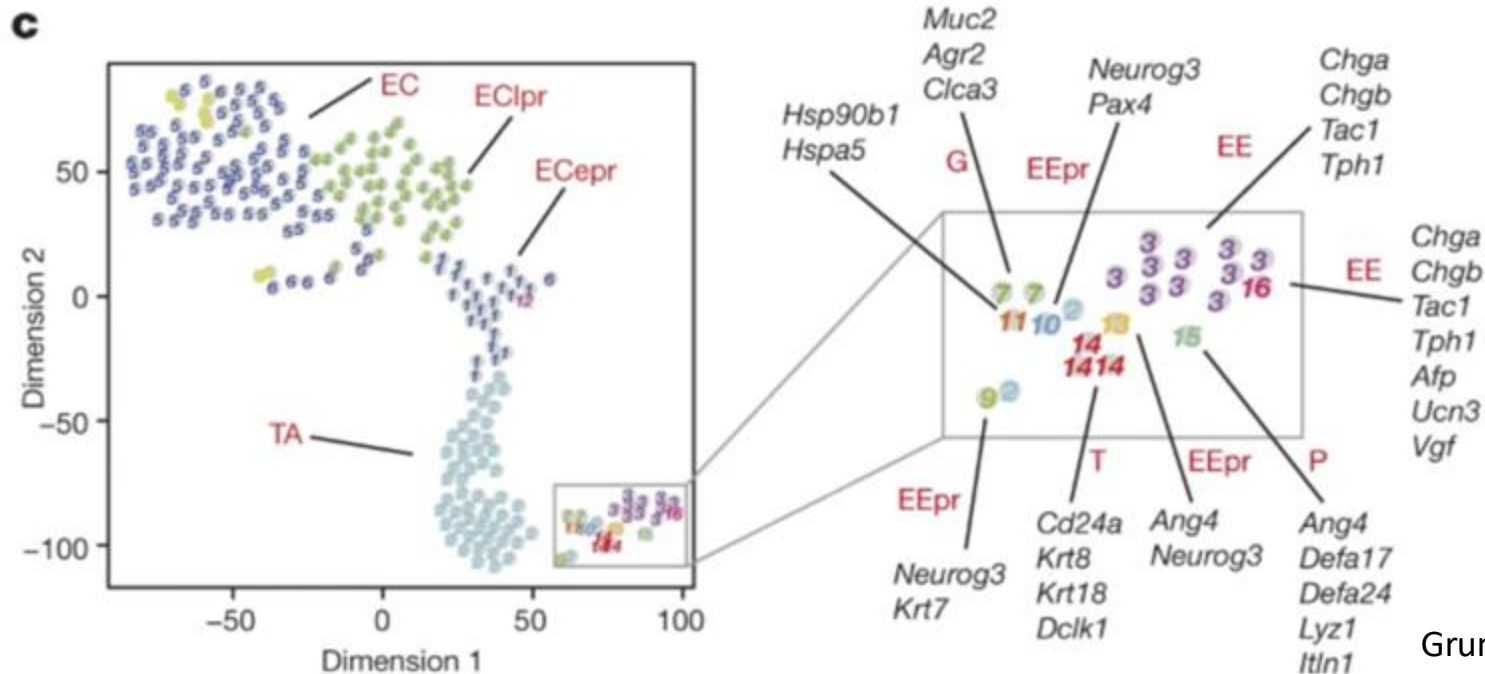
Silhouette analysis for KMeans clustering on sample data with $n_clusters = 6$



Clustering granularity



RaceID identifies rare cell types



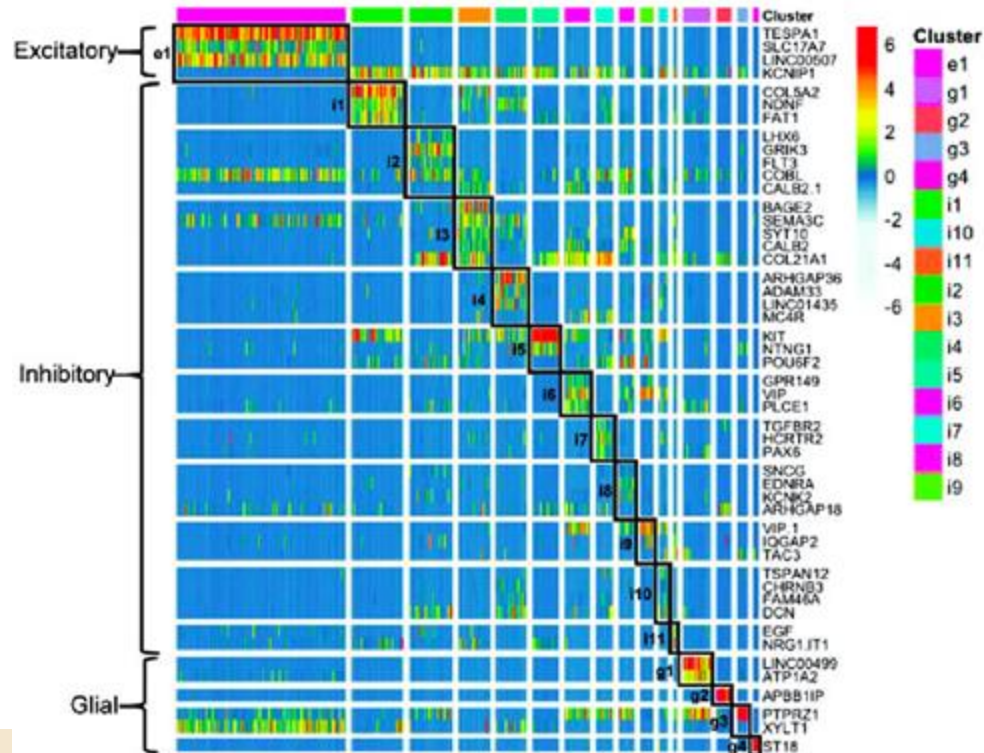
Grun et al, Nat
Methods, 2015

To think about when clustering

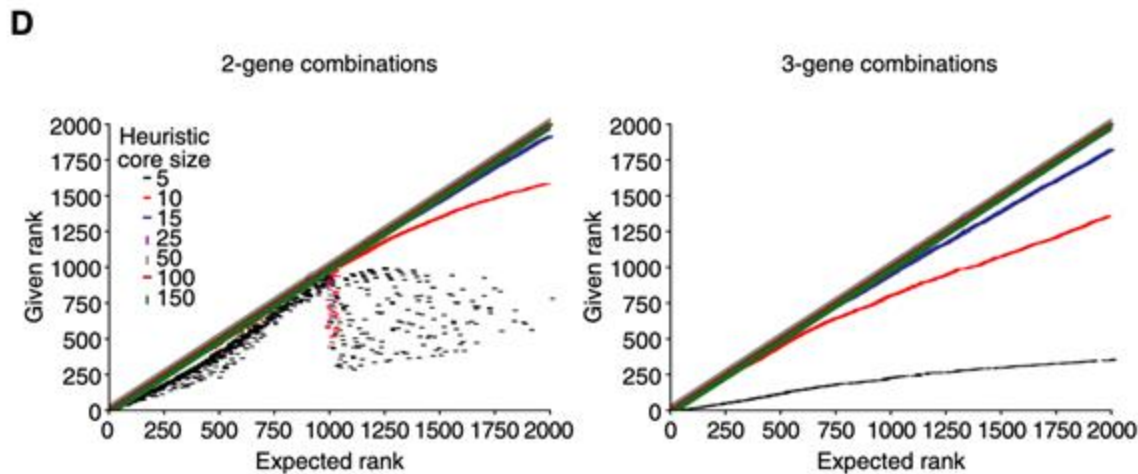
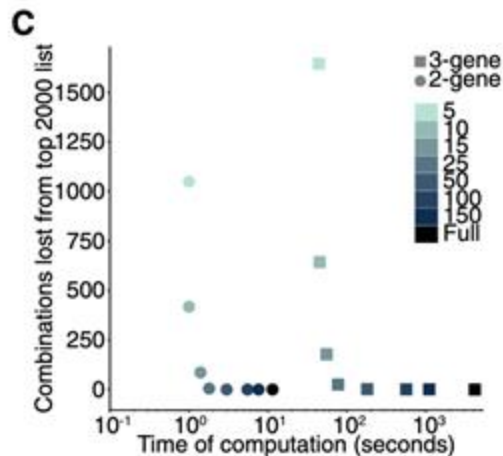
- Clustering works well, even with very sparse data
- Mathematical optimum vs biological interpretability
- Annotation is often the biggest bottleneck



Marker genes identify clusters

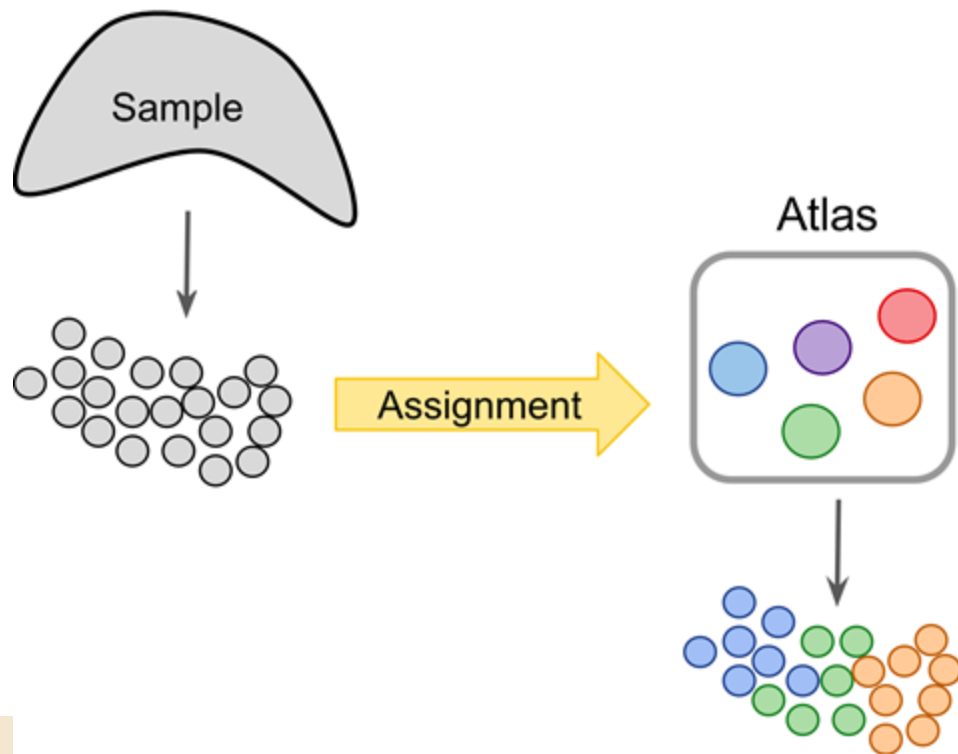


Comet uses combinatorial approach

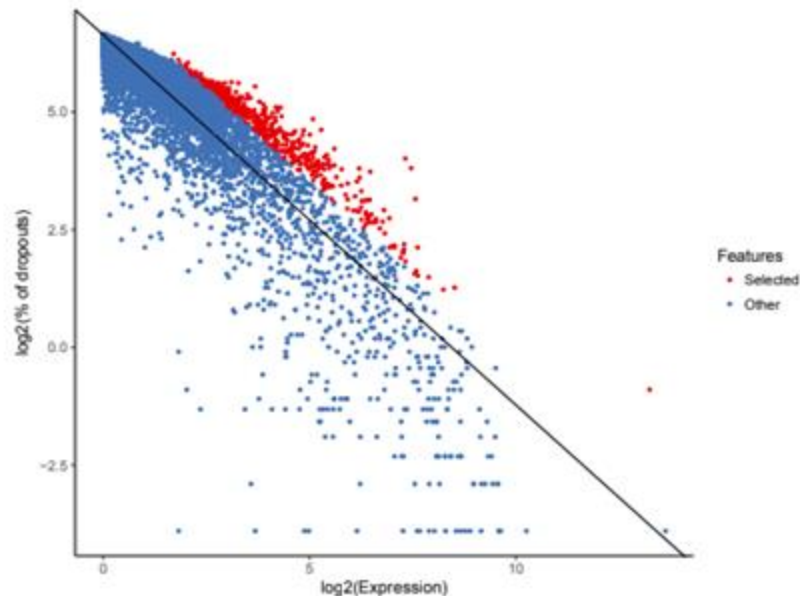
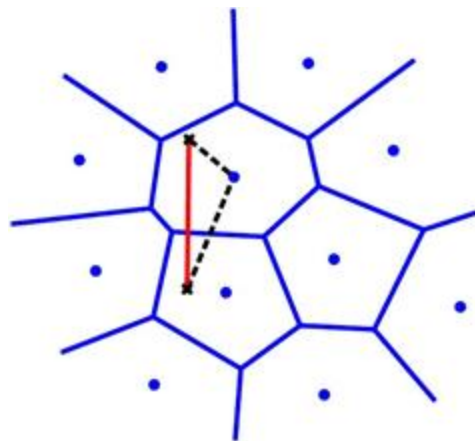


Delaney et al,
MSB, 2019

Comparisons across datasets

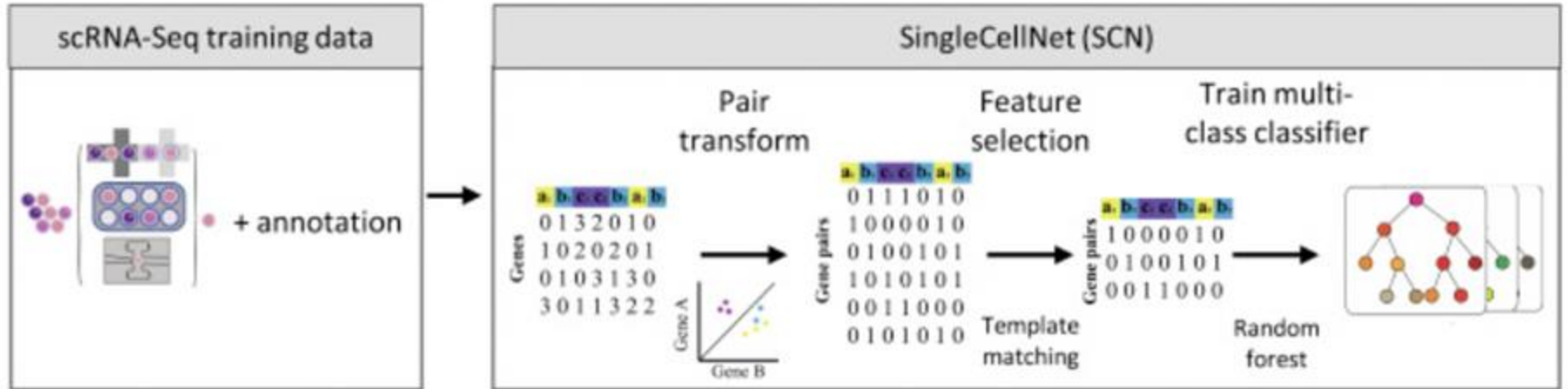


Scmap finds nearest cluster



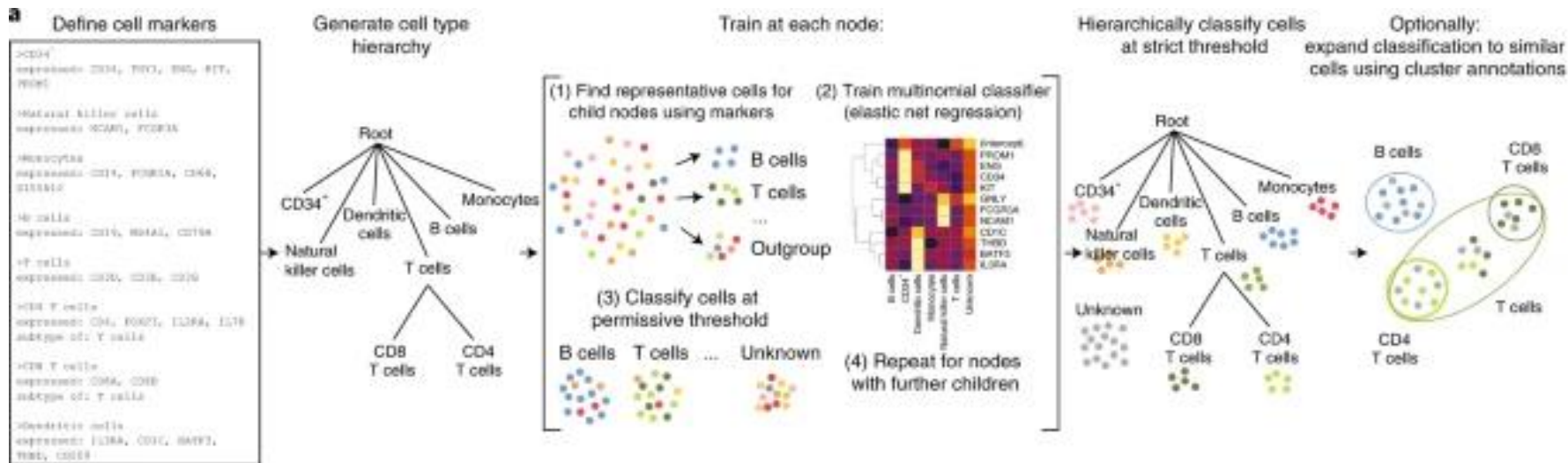
Kiselev et al, Nat
Methods, 2018

SingleCellNet uses Random Forests



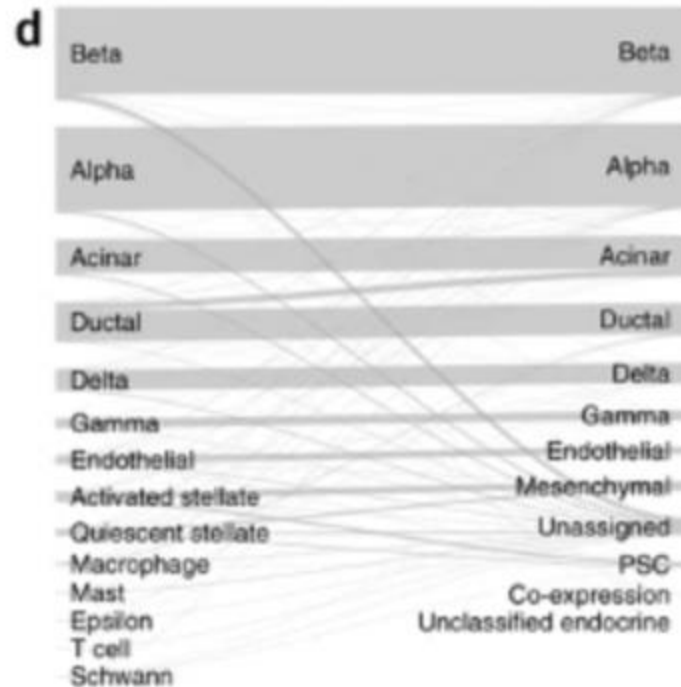
Tan and Cahan, Cell
Systems, 2019

Garnett does supervised projection

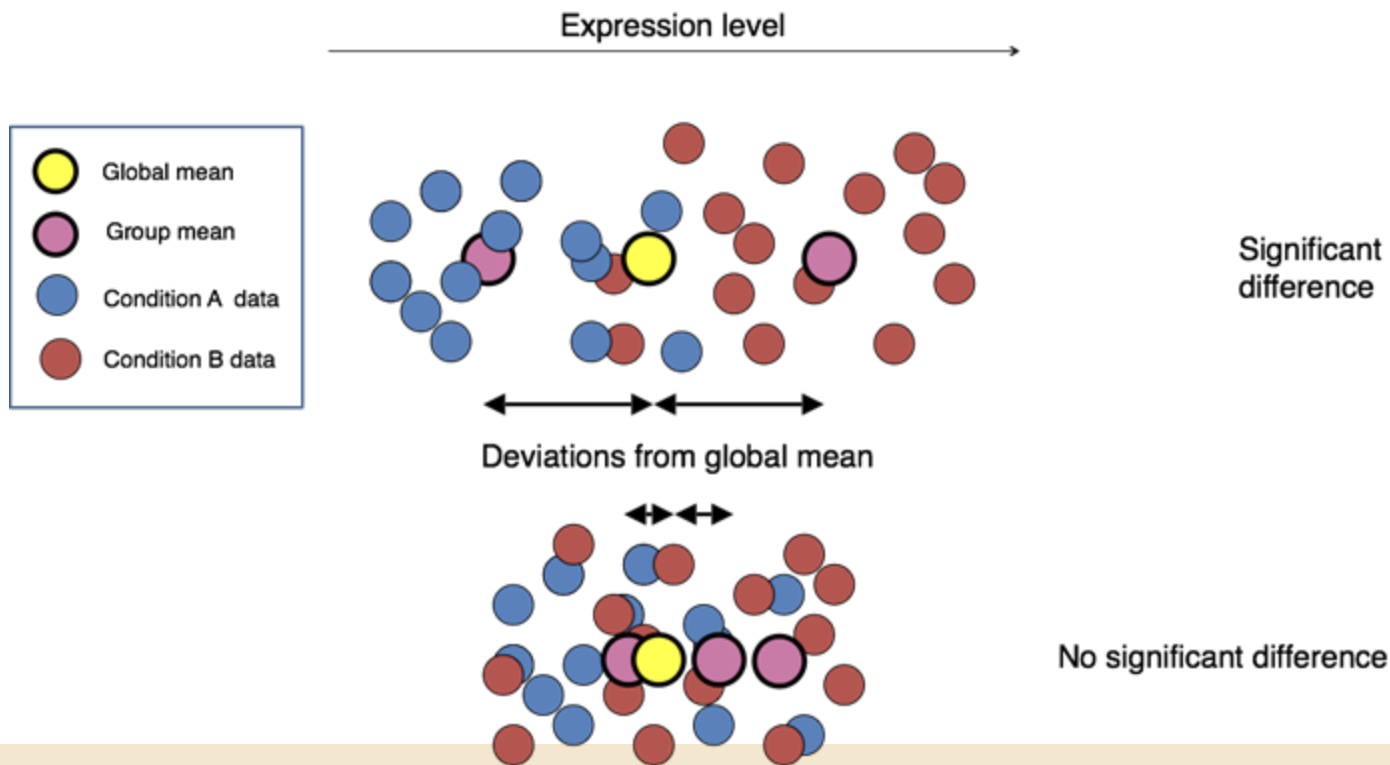


Pilner et al, Nat
Methods, 2019

Sankey diagrams visualize mappings



Differential expression

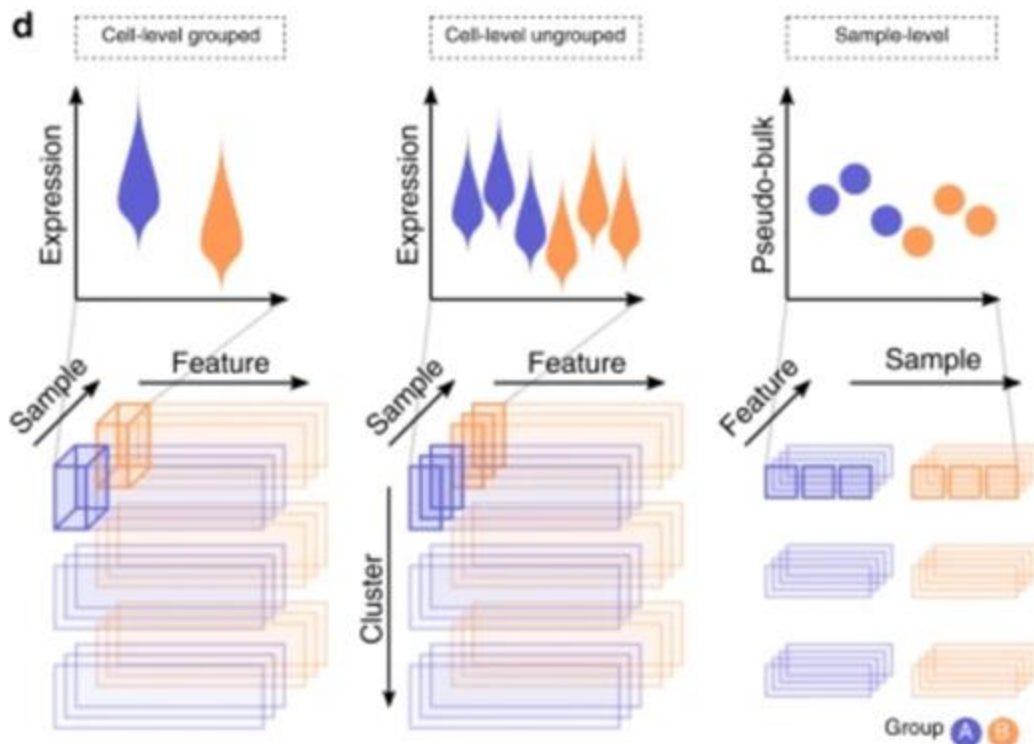


Non-parametric tests perform well



Sonesson and Robinson,
Nat Methods, 2018

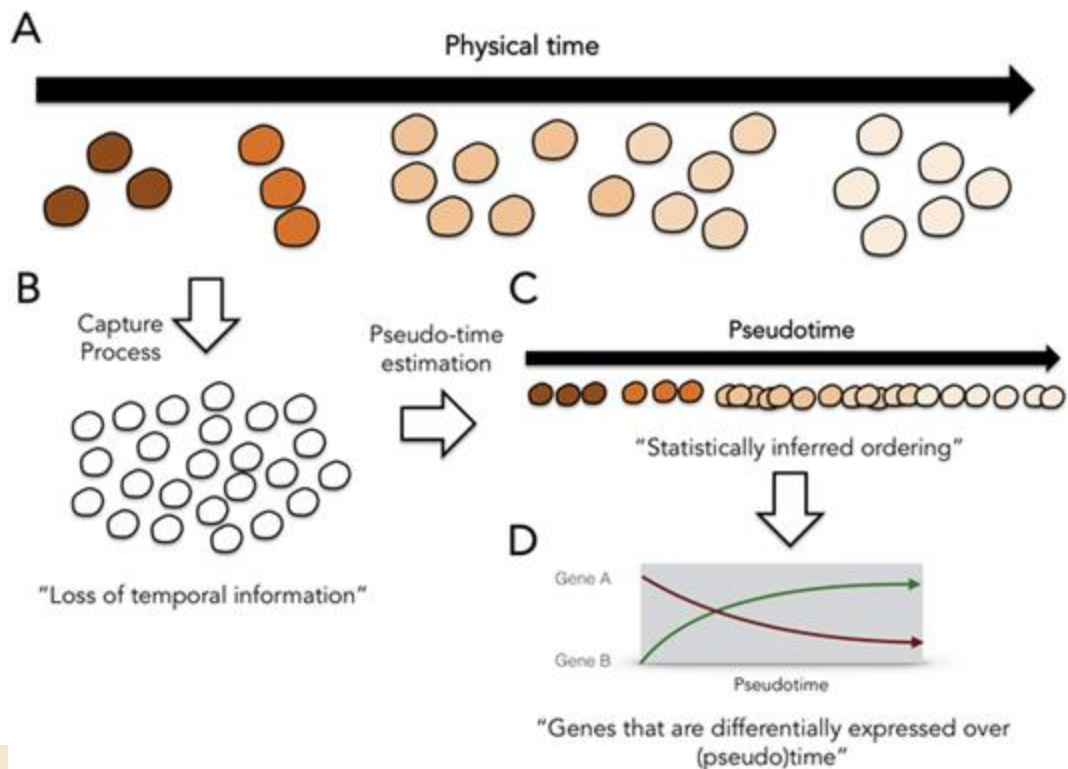
Muscat allows for complex setups



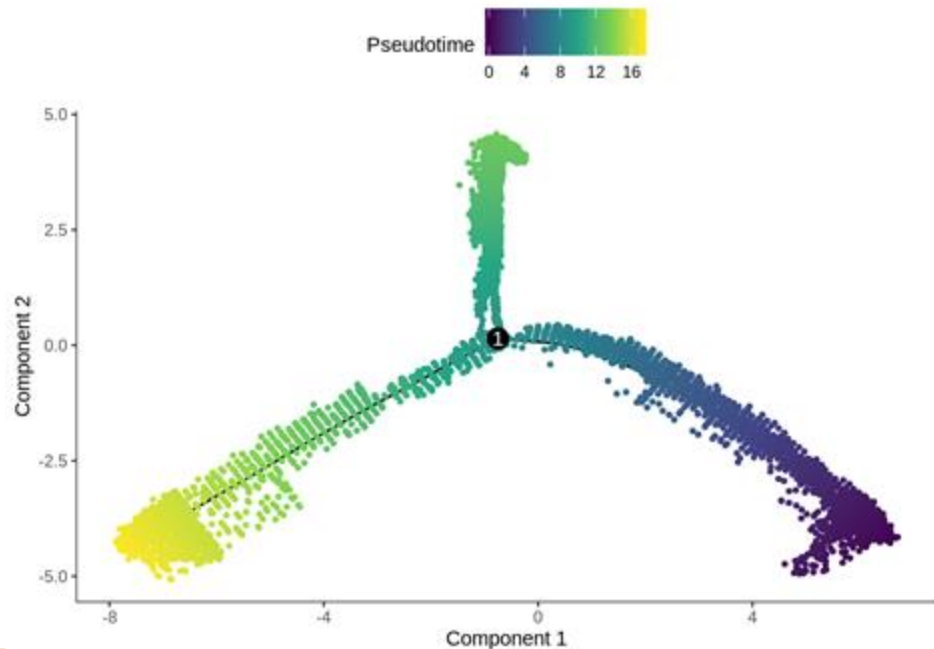
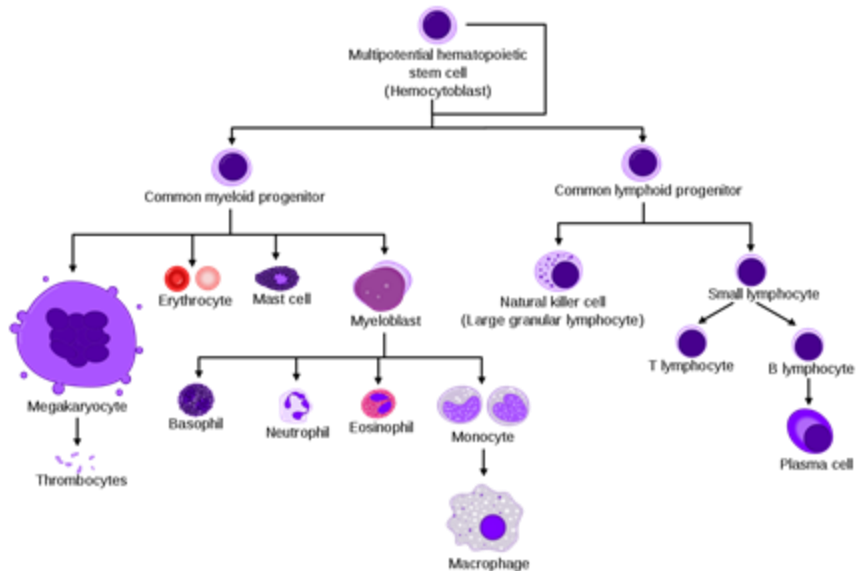
Crowell et al, Nat
Communications, 2020

Pseudotime and RNA velocity

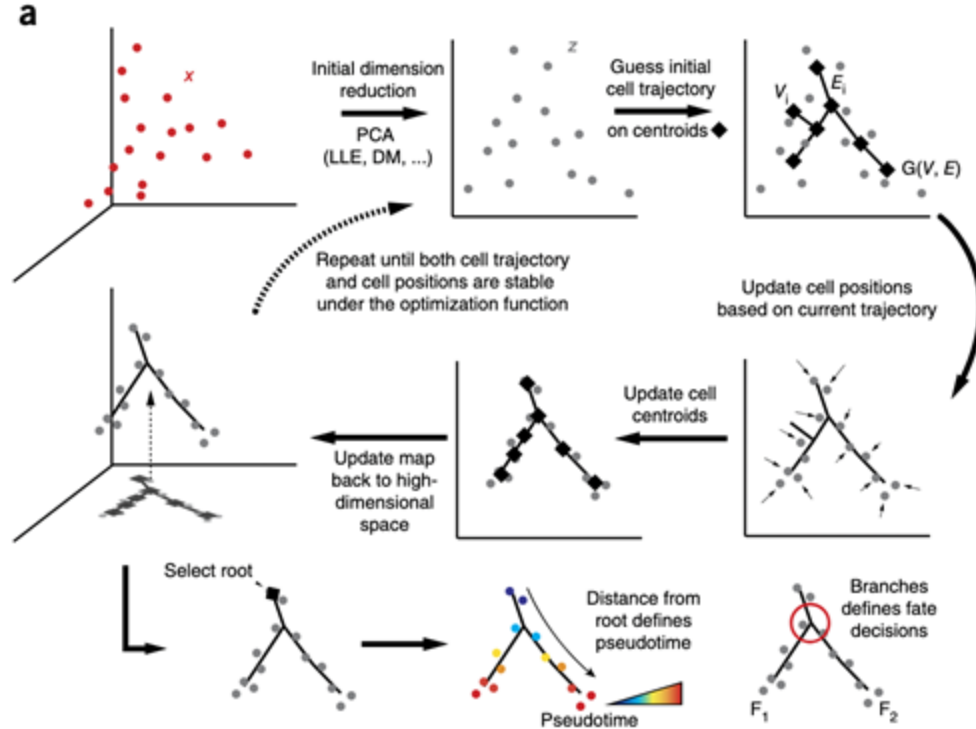
Pseudotime



Branching

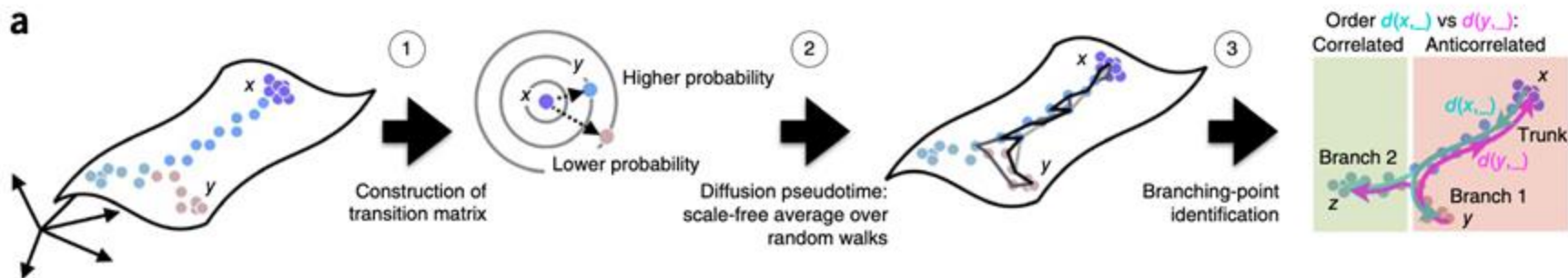


Reverse graph embedding with Monocle²



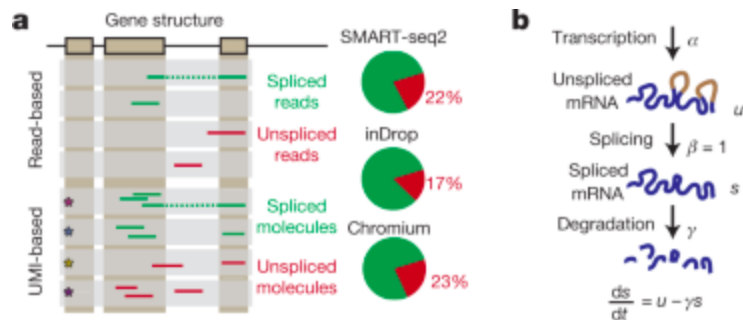
Qiu et al, Nat
Methods, 2019

Diffusion pseudotime



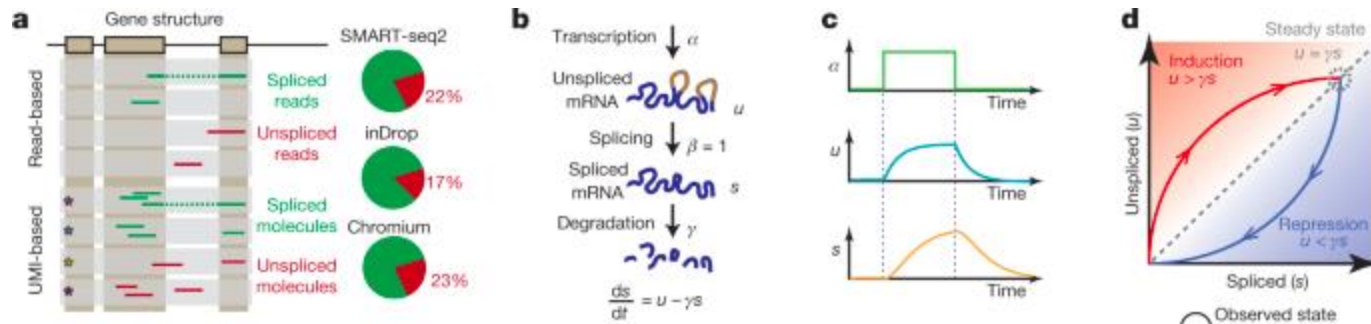
Haghverdi et al, Nat
Methods, 2016

RNA velocity

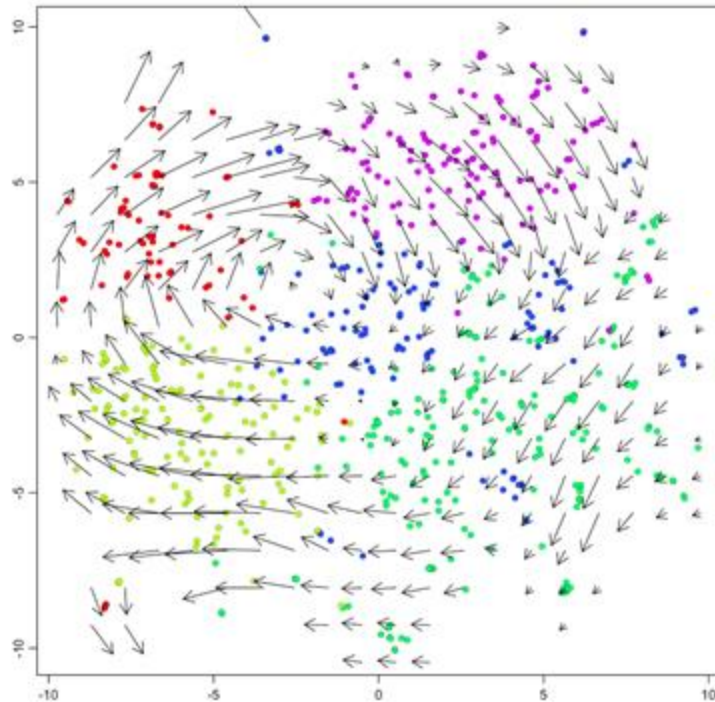


La Manno et al,
Nature, 2018

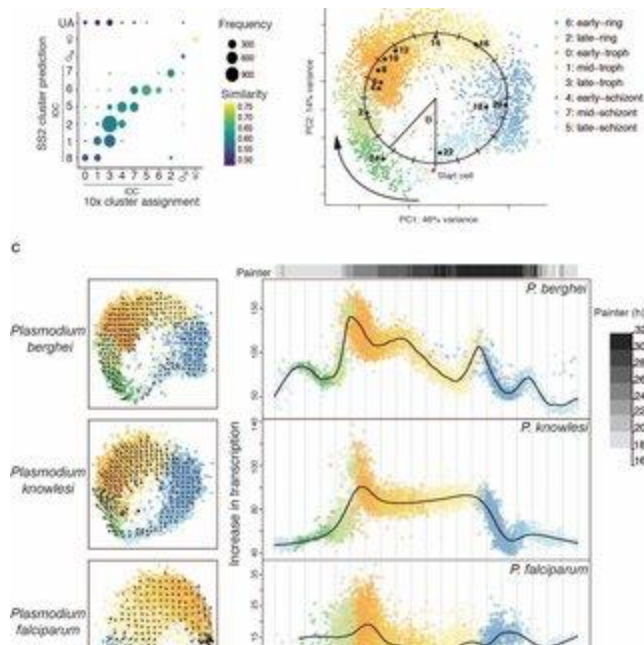
RNA velocity



RNA velocity and cell transitions



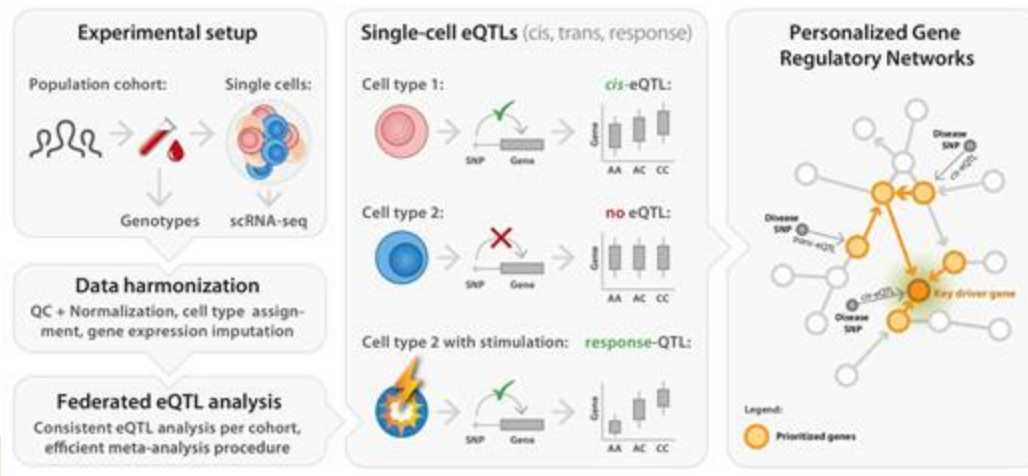
RNA velocity and real time



Howick et al,
Science, 2019

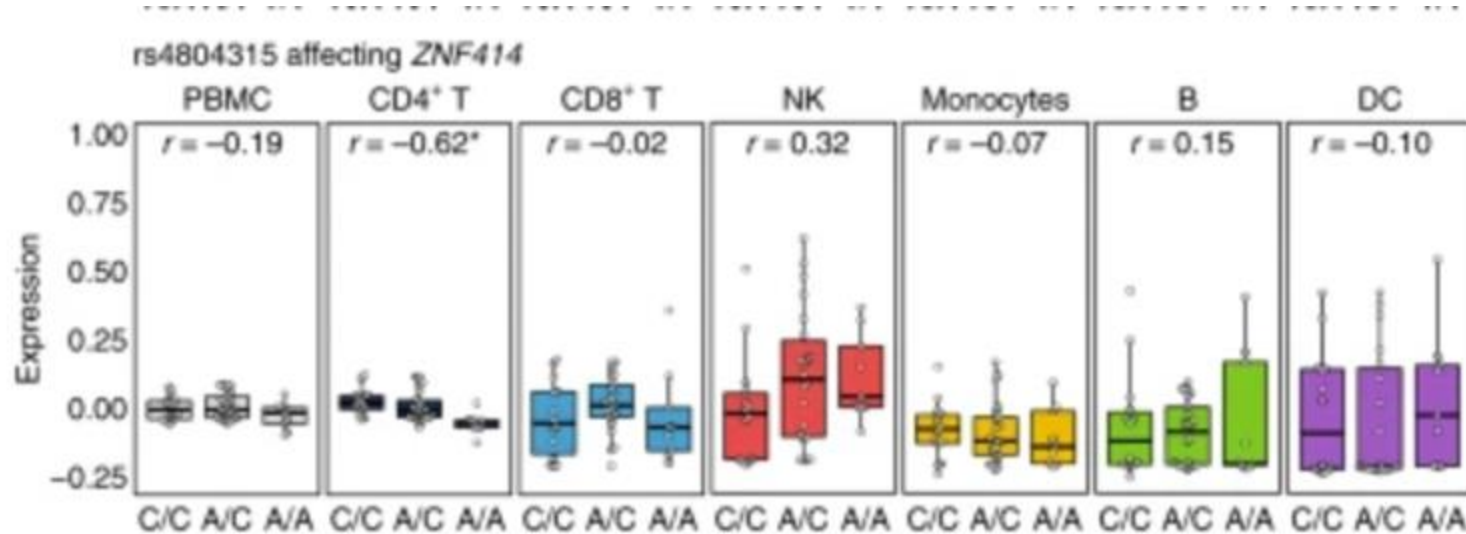
Clustering by genotype and CITE-seq

Single-cell eQTLs



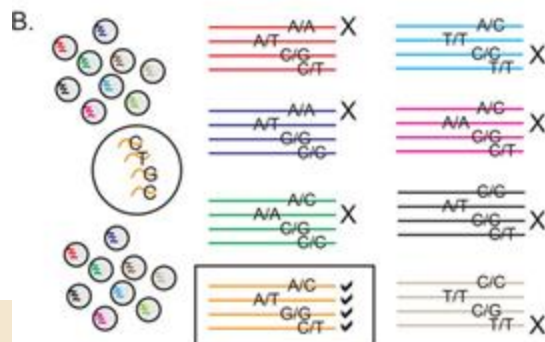
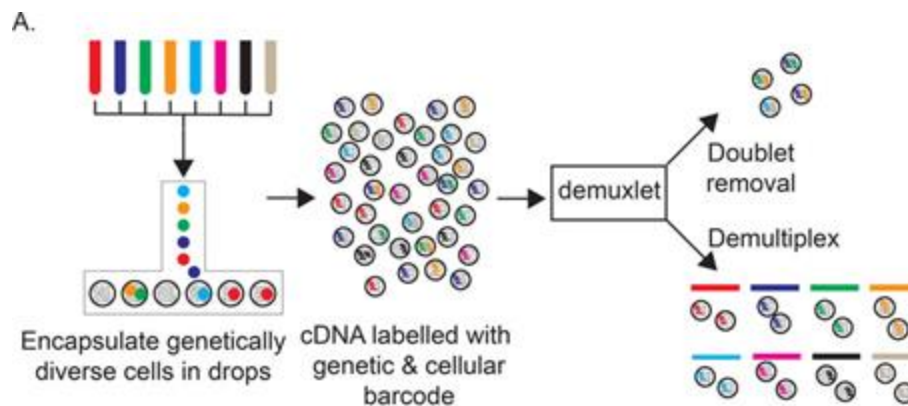
van der Wijst et al,
eLife, 2018

sceQTLs for PBMCs from 45 donors

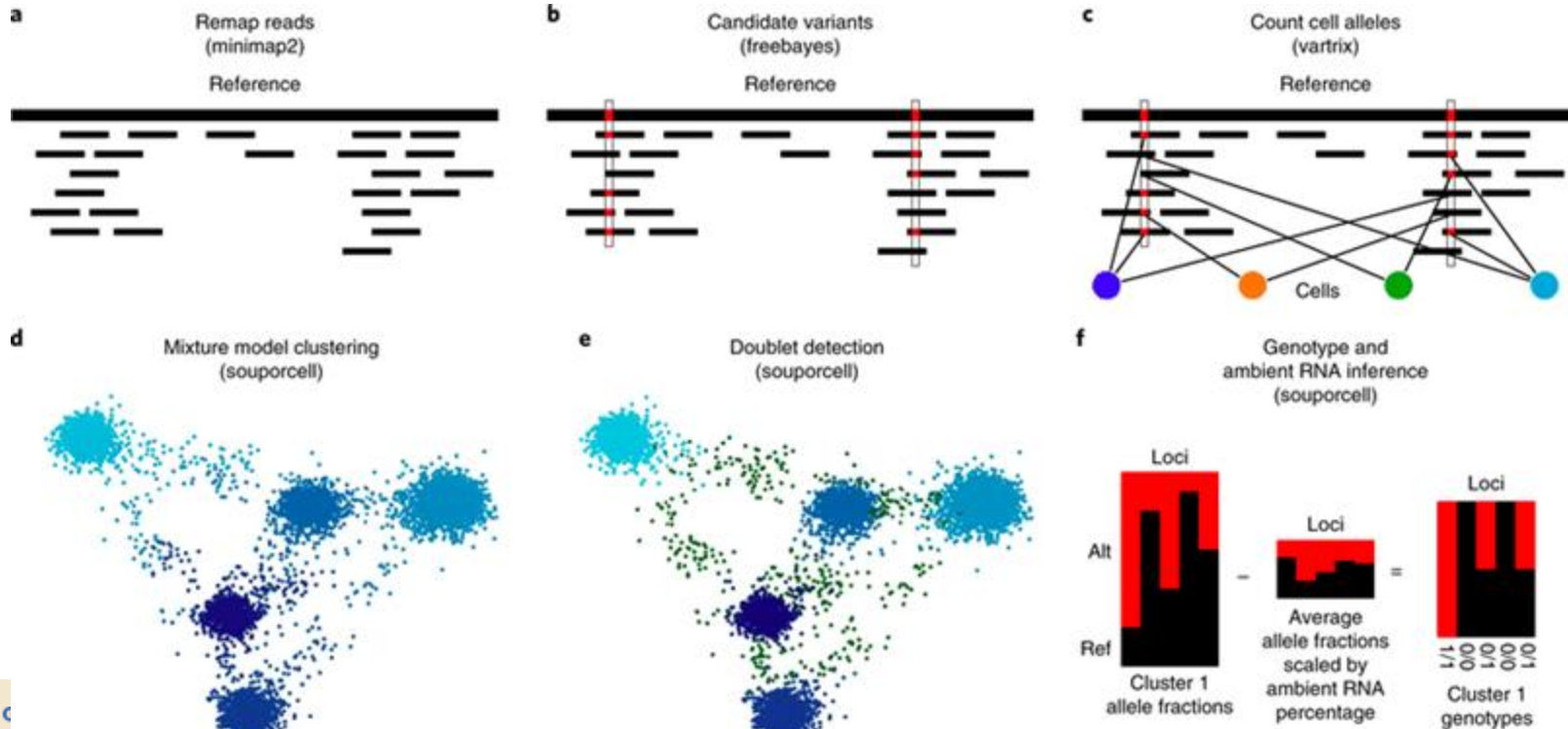


van der Wijst et al,
Nat Genetics, 2018

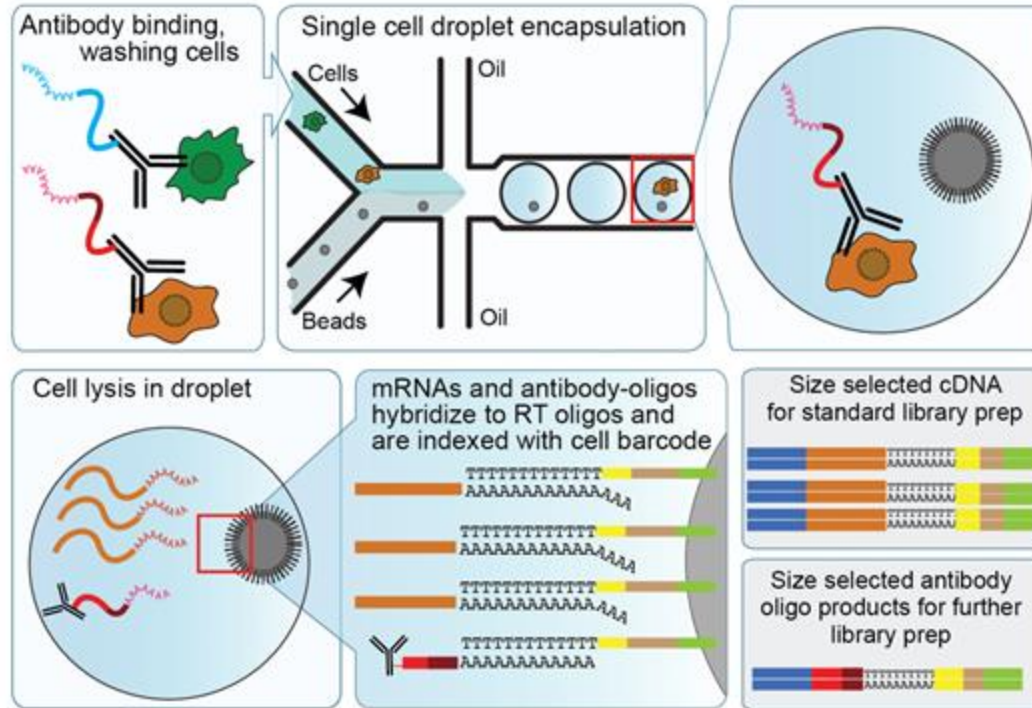
Demuxlet clusters by genotype



Souporcell clusters without reference

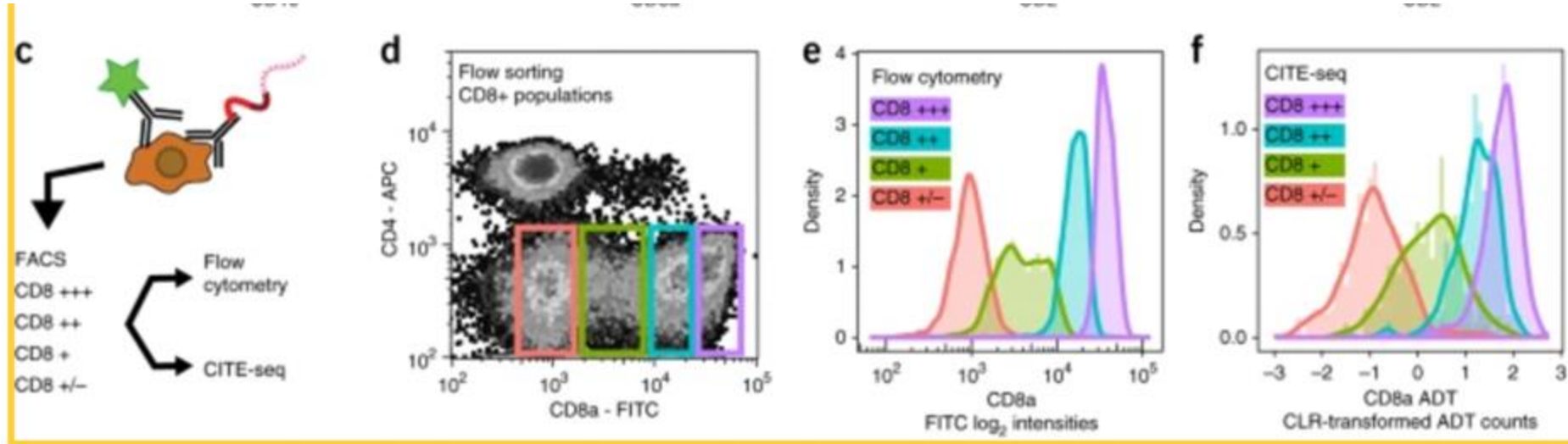


CITE-seq measures proteins

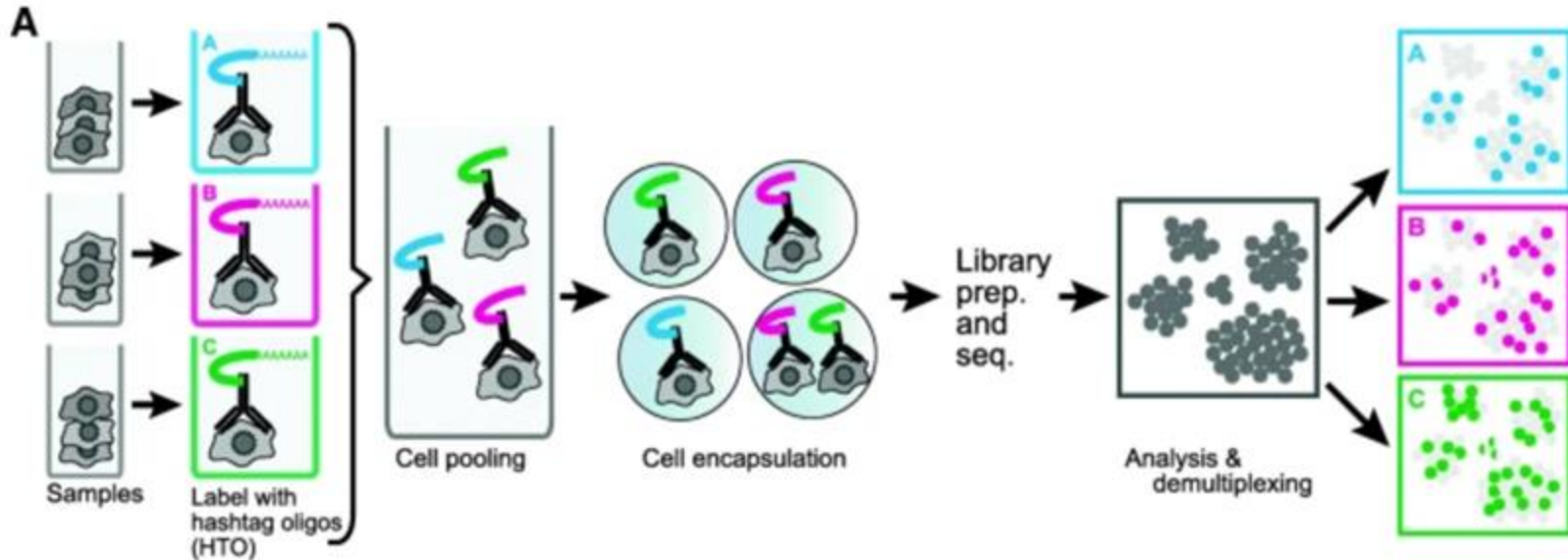


Stoeckius et al, Nat
Methods, 2017

CITE-seq similar to flow cytometry



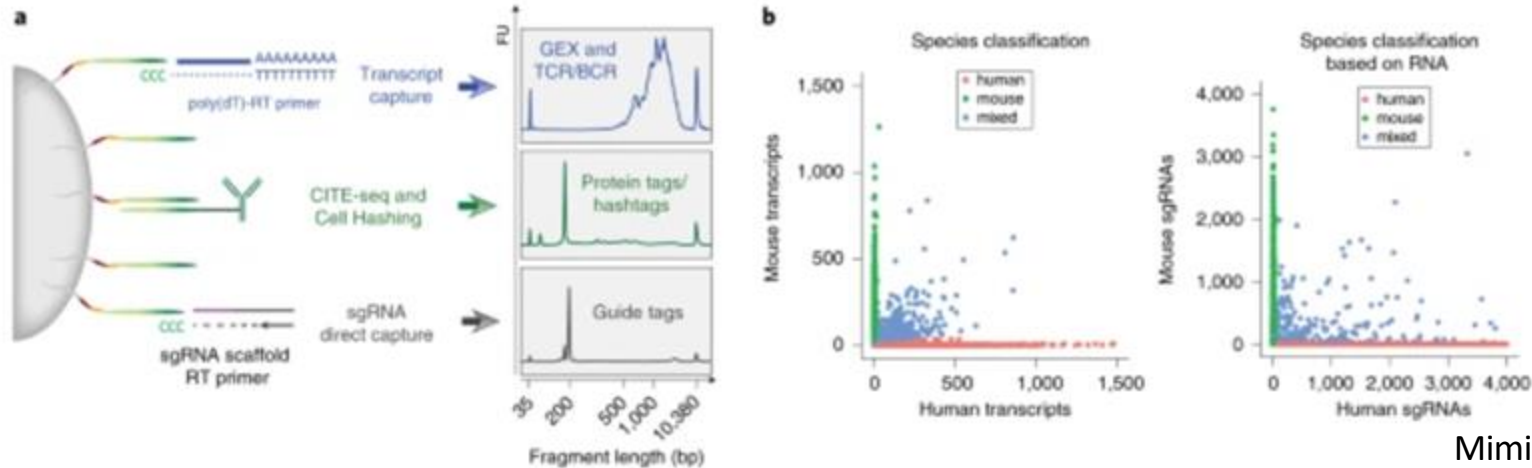
Demultiplex using cell hashing



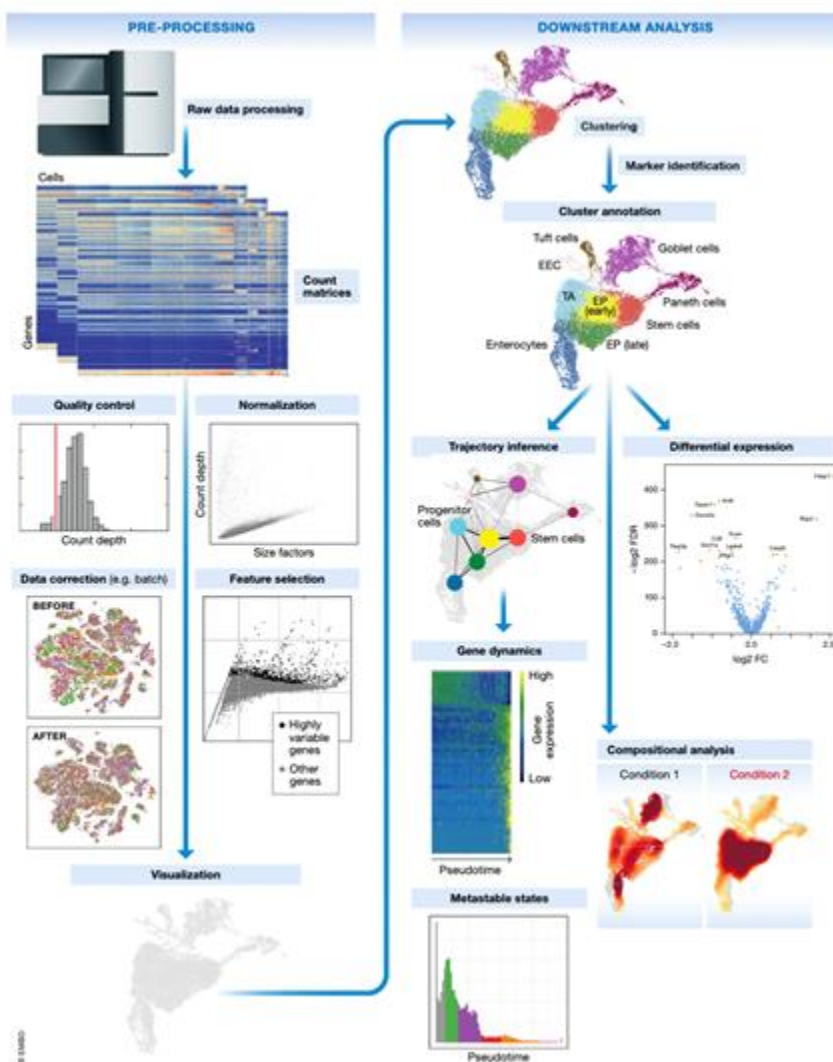
Stoeckius et al,
Genome Biology, 2018

ECCITE-seq allows multi-omics

Fig. 1: ECCITE-seq allows simultaneous detection of transcriptome, proteins, clonotypes and CRISPR perturbations.



Mimitou et al, Nat
Methods, 2019

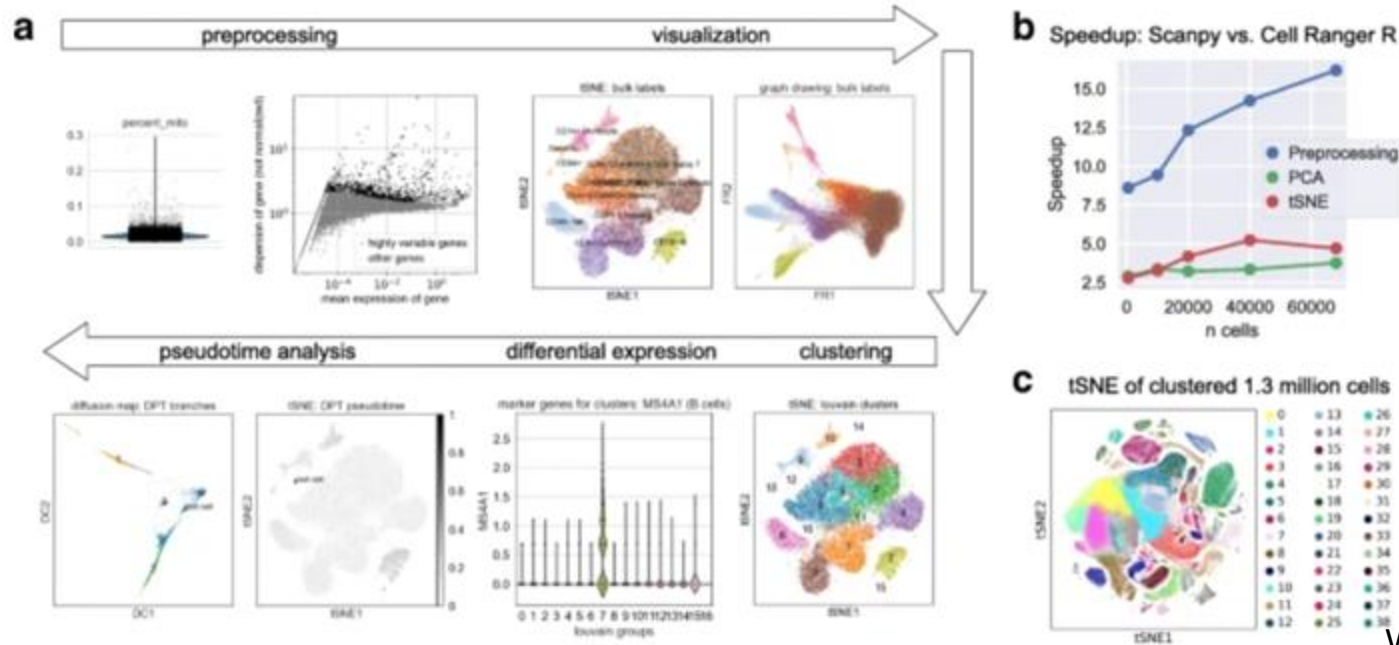


Luecken and Theis,
MSB, 2019

Seurat is a single cell toolkit



Scanpy supports analysis in python



Wolf et al, Genome Biology, 2018

R vs python



JRME-CLARK.TUMBLR